A Comprehensive Survey of Retrieval-Augmented Generation (RAG) Evaluation and Benchmarks: Perspectives from Information Retrieval and LLM

IBRAHIM AL AZHER, Northern Illinois University, USA
HAMED ALHOORI, Northern Illinois University, USA

Retrieval-Augmented Generation (RAG) integrates information retrieval with large language models (LLMs) to produce grounded and factually consistent outputs. RAG addresses critical limitations of standalone LLMs by enriching parametric knowledge with external evidence, notably reducing hallucinations and improving factual grounding. However, evaluating RAG remains particularly challenging because it involves multiple interdependent components—retrievers, generators, and their combined end-to-end pipeline—where errors in one module can propagate and distort overall performance. Rigorous RAG evaluation is essential to ensure reliability, fairness, and robustness, especially as these systems are increasingly deployed in knowledge-intensive and high-stakes domains. In this survey, we examine evaluation approaches across three dimensions: (1) Retriever evaluation, drawing from traditional information retrieval practices to assess relevance, coverage, and efficiency; (2) Generator evaluation, leveraging metrics such as fluency, factuality, and LLM-as-a-Judge frameworks to measure textual quality and grounding; and (3) End-to-end evaluation, which captures how retrieval and generation interact in real-world RAG systems. We synthesize insights on system performance, compile and categorize key benchmark datasets for retriever-centric, generator-centric, and end-to-end systems, highlight cross-cutting challenges such as bias and resource constraints, and analyze current limitations. Finally, we outline best-practice guidelines and future research directions, including unified evaluation frameworks, bias mitigation strategies, and the development of adaptive, task-aware benchmarks. Our goal is to provide a roadmap for building reliable and trustworthy RAG evaluation methodologies that can support the next generation of retrieval-grounded LLM applications.

CCS Concepts: • **Information systems** → **Retrieval models and ranking**; • **Computing methodologies** → *Natural language processing*.

Additional Key Words and Phrases: Retrieval Augmented Generation, Evaluation, Retievar, Generator, LLM, Information Retrieval

1 Introduction

Large Language Models (LLMs) [1, 178] have emerged as a transformative technology, demonstrating remarkable capabilities in a wide array of tasks, from generating fluent text and creative imagery to writing complex code. Their power lies in their vast parametric knowledge, encoded within billions of network weights during pre-training. However, this reliance on internal knowledge is also a significant weakness. LLMs are prone to "hallucination," fabricating facts when information is absent from their training data [78]. Furthermore, their knowledge is static, lacking access to real-time events or domain-specific information crucial for specialized fields like healthcare, finance, or corporate policy. These limitations hinder their reliability in knowledge-intensive, real-world applications.

To overcome these challenges, Retrieval-Augmented Generation (RAG) has been introduced as a powerful paradigm [100]. RAG enhances LLMs by integrating an external information retrieval component. Instead of relying solely on its internal knowledge, a RAG system first queries an external knowledge base, such as a vector database, to fetch relevant, up-to-date documents. This retrieved context is then provided to the LLM's generator, which synthesizes a final answer grounded in this external, non-parametric knowledge. This dual-component architecture significantly improves factual accuracy, reduces hallucinations, and enables the model to provide answers on niche topics, making LLMs more trustworthy and applicable.

Authors' Contact Information: Ibrahim Al Azher, Northern Illinois University, DeKalb, IL, USA, iazher1@niu.edu; Hamed Alhoori, Northern Illinois University, DeKalb, IL, USA, alhoori@niu.edu.

As RAG systems become increasingly integral to modern AI applications, the need for robust and comprehensive evaluation methodologies becomes paramount. Evaluation is the cornerstone of model development, providing the necessary feedback to measure performance, compare different architectures, and identify areas for improvement. However, evaluating a RAG system is inherently complex due to its multi-stage pipeline. A thorough assessment must consider the distinct performance of the retriever, the quality of the generator, and the synergistic effectiveness of the end-to-end system. Without a proper evaluation framework that addresses each of these facets, building and optimizing effective RAG systems remains a significant challenge.

While initial evaluation frameworks such as RAGAs [44] and ARES [158] have laid important groundwork, a holistic approach is still needed. The multifaceted nature of RAG demands a diverse set of metrics. The retriever's performance, for instance, can be measured using traditional information retrieval metrics like precision and recall. In contrast, the generator's output requires assessment through a combination of classic NLP-based metrics for fluency and the increasingly popular LLM-as-a-judge [220] approach for semantic and factual correctness. Furthermore, a complete evaluation must also incorporate system-level metrics, such as latency and computational efficiency, which are critical for practical deployment.

We systematically review and synthesize findings from over **220** papers to provide a holistic overview of the RAG evaluation landscape in this comprehensive survey. We delve into the spectrum of evaluation methodologies for each component of the RAG pipeline, detailing both the underlying text retrieval methods and the traditional Information Retrieval metrics for assessing the retriever, a range of NLP-based metrics, and the emerging "LLM-as-a-Judge" paradigm for measuring the generator's performance, as well as holistic metrics for the end-to-end system. Our work synthesizes optimal methods and best practices for improving the performance of both the retrieval and generation components. Additionally, we provide an overview of key benchmark datasets, categorizing them by their utility for evaluating the retriever, the generator, and the complete end-to-end system. Critically, this survey also identifies the current drawbacks and limitations in existing RAG evaluation methods (retriever, generator based) and proposes promising directions for future research to build more reliable, efficient, and effective RAG evaluation systems.

This survey systematically explores the landscape of RAG evaluation. We begin in Section 2 by reviewing related works to position our survey within the existing literature. The technical discussion starts in Section 3 by detailing various methods for text retrieval. We then delve into the core evaluation methodologies, examining specific metrics and frameworks for the retriever (Section 4), the generator (Section 5), and the complete end-to-end system (Section 6). To ground these concepts, Section 7 reviews key benchmark datasets for Retriever centric, generator centric and end to end system. Finally, we address the current challenges and limitations of RAG component based evalaution in Section 8 and conclude by outlining best practices and future research directions for evaluation in Section 9.

2 Related Work

Evaluating Retrieval-Augmented Generation (RAG) systems is a critical and evolving field. While numerous studies [7, 79, 217] have examined the architecture and application of RAG, a smaller, more recent body of work focuses specifically on its evaluation [19, 55, 206]. This section reviews the existing literature, first by covering broad surveys of RAG and then by delving into specific evaluation frameworks, metrics, and benchmarks.

Numerous surveys provide broad overviews of RAG architectures and applications from diverse perspectives. A comprehensive analysis of retrieval–generation integration in knowledge-intensive tasks addresses key challenges and model robustness [67], while a systematic review explores the convergence of RAG and large language models through architectural, training, and application lenses [49]. Other works include an application-focused study emphasizing RAG

deployments [7], and in-depth work detailing LLM interactions with modules for query rewriting, retrieval, reranking, and reading [225]. Further contributions have presented a taxonomy of RAG and Retrieval-Augmented Understanding (RAU) components [75], and an engineering-oriented overview of RAG for AI-generated content [217]. From a retrieval-centric standpoint, another survey divides RAG into pre-retrieval, retrieval, post-retrieval, and generation phases [79]. For structured data, the GraphRAG framework has also been surveyed, covering indexing, graph-based retrieval, and generation [142]. Although these efforts have enriched the understanding of RAG, dedicated evaluation-focused surveys remain scarce.

The evaluation of a Retrieval-Augmented Generation (RAG) system is a multi-faceted process, with a diverse array of metrics that can be categorized by their focus on the retriever, the generator, or the end-to-end pipeline. For retriever-focused evaluation, performance is typically assessed using traditional, module-wise information retrieval metrics such as precision, recall, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (nDCG) to measure the quality of the ranked documents [58, 225]. For generator-focused evaluation, the quality of the final text is measured using classic text-overlap metrics like BLEU and ROUGE, broader categories of human-centric and machine-learned metrics [23], and, more recently, the scalable "LLM-as-a-Judge" paradigm [65]. Finally, end-to-end evaluation assesses the synergistic performance of the entire system, using both broad measures that distinguish between unlabeled-content and labeled-content assessments [58] and more granular frameworks like RAGAS that evaluate faithfulness and relevance [44], or specialized benchmarks like RGB [28], CRUD [119], and RECALL [114] that test for noise robustness, update capabilities, and misinformation resilience, respectively.

While comprehensive surveys on RAG evaluation are limited, several key studies have proposed specific frameworks and benchmarks. Such as, overview of RAG evaluatin methodologies by datasets, retrievers, indexing and databases [19], survey on RAG specific dataset and evaluations [55]. Moreovere, a study proposed a unified evaluation process named Auepora, which centers on three fundamental questions—What to Evaluate?, How to Evaluate?, and How to Measure? By systematically mapping evaluation targets, datasets, and metrics, Auepora offers a clear and accessible methodology for benchmarking RAG systems across both retrieval and generation aspects. Other specialized frameworks have also been developed. Such as, MIRAGE was introduced as a benchmark with over 7,600 questions from medical QA datasets [196] to evaluate medical RAG systems. Recognizing that traditional end-to-end evaluation is computationally expensive, Salemi et al. [161] proposed eRAG, which derives document-level relevance labels more efficiently and shows a stronger correlation with downstream performance. Meanwhile, Gan et al. [55] systematically review evaluation approaches for system performance, factual accuracy, safety, and computational efficiency.

Despite these valuable contributions, a detailed survey focused exclusively and comprehensively on RAG evaluation remains a notable gap in the literature. This survey addresses this gap by providing a holistic review of the entire RAG evaluation pipeline. We offer a deep dive into retriever evaluation, covering not only traditional Information Retrieval (IR) metrics but also the underlying text retrieval approaches themselves. For the generator, we analyze a spectrum of assessment techniques, from classic NLP-based metrics to the modern "LLM-as-a-Judge" paradigm, while also covering end-to-end frameworks that measure the synergistic performance of the whole system. Across all three components, we compile relevant benchmark datasets, analyze system performance metrics like latency, detail the current shortcomings of existing methods, and propose concrete future directions to overcome these limitations and build more reliable and effective RAG evaluation frameworks.

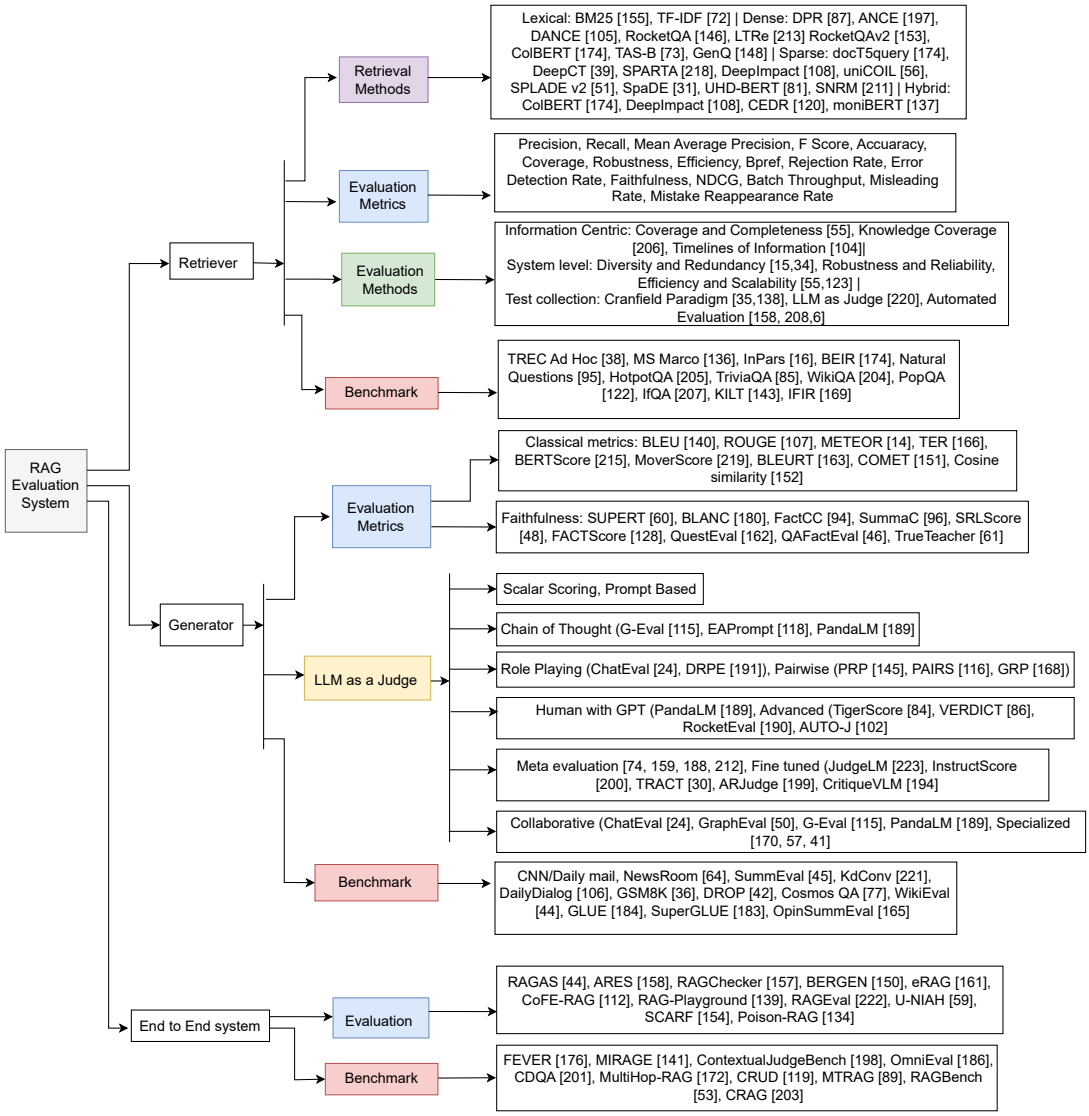


Fig. 1. A flowchart of the RAG evaluation process, detailing the methods, metrics, and benchmarks for the retriever, generator, and end-to-end system.

3 Retriever Methods

The performance of a Retrieval-Augmented Generation (RAG) system is dependent on its Retrieval component. This crucial module is tasked with finding relevant documents or passages from a large knowledge base in response to a user's query, which are then used to ground the Large Language Model's (LLM) generation. The effectiveness of this retrieval step directly influences the factual accuracy and relevance of the final output, making its evaluation essential.

Therefore, understanding the underlying mechanisms and trade-offs of different retrieval methods is a prerequisite for effective evaluation, as the choice of architecture directly informs which performance metrics are most relevant. To provide a comprehensive overview, this survey details the evolution of these architectures. We will examine a spectrum of techniques for text retrieval from the sources, including foundational Lexical Statistical models, modern sparse retrieval and dense retrieval methods, advanced hybrid and multi-vector approaches, and sophisticated strategies for context-aware and personalized retrieval.

3.1 Lexical and Statistical Models

BM25 [155] and TF-IDF [72] are classical term-weighting models that rely on exact-term overlap between queries and documents, and they are often used as strong baselines in information retrieval tasks. However, these models struggle to capture semantic similarities when there is no lexical overlap. To address this limitation, Generalized Language Models (GLM) extend traditional language model ranking by incorporating word embedding similarities, enabling them to model term dependencies and support semantic matching. In addition, model-based feedback methods offer a more principled approach to query refinement by updating the query language model based on feedback documents, rather than relying on simple term expansion. This ensures that the revised query remains probabilistically consistent while improving retrieval effectiveness.

3.2 Dense Retrieval

Modern information retrieval has shifted from traditional lexical matching to dense retrieval, which represents queries and documents as dense vectors (embeddings) in a shared semantic space. This enables conceptual matching beyond exact keyword overlap, fundamentally changing how relevance is computed. The initial breakthrough in dense retrieval was the dual-encoder architecture, exemplified by Dense Passage Retrieval (DPR) [87]. DPR uses two independent BERT-based encoders to embed queries and passages into a low-dimensional vector space where retrieval is performed via efficient nearest-neighbor search. Trained using "in-batch negatives," DPR was one of the first models to consistently outperform the strong lexical baseline BM25 in top-k retrieval accuracy. However, a key limitation of its training is that in-batch negatives are often not challenging enough to teach fine-grained distinctions.

Subsequent research focused heavily on improving negative sampling strategies. For example, Approximate Nearest-neighbor negative Contrastive Learning (ANCE) [197] introduced sampling "hard negatives" from the entire corpus using an asynchronously updated ANN index, which accelerates convergence. Building directly on this, DANCE [105] proposed a dual contrastive learning paradigm to create a more uniform embedding space and achieve more balanced recall. Evaluated on the widely used MS MARCO dataset, DANCE showed significant retrieval performance improvements over ANCE on metrics like NDCG@10 and MRR@100. Moreover, RocketQA [146] further strengthened training by incorporating cross-batch negatives and a sophisticated "denoised" hard negative filtering technique using a cross-encoder.

Another critical challenge is the mismatch between training objectives and the final inference task. Learning To Retrieve (LTRe) [213] directly addresses this by training models to retrieve from the full document collection, aligning the training process with real-world inference. To bridge the gap between retrieval and re-ranking, RocketQAv2 [153] pioneers a dynamic listwise distillation strategy that enables the joint training of the retriever and a more powerful re-ranker. Architecturally, “late interaction" models like ColBERT [174] provide finer-grained relevance scoring than standard bi-encoders while remaining much faster than full cross-encoders. This concept is leveraged by TAS-B [73], which enhances the bi-encoder by using dual supervision from both a cross-encoder and a ColBERT model. Finally,

to adapt dense retrievers to new domains, GenQ [148] introduces an unsupervised approach that generates synthetic query-passage training pairs.

3.3 Sparse Retrieval

While dense retrieval models leverage embeddings for semantic matching, sparse retrieval methods combine the semantic power of neural representations with the efficiency of traditional inverted indexes. This approach seeks to bridge the “lexical gap," where queries and documents use different words to describe the same concept. A prominent document expansion technique is docT5query [174], which uses a T5 model to generate and append synthetic questions to documents, making them more "keyword rich" for traditional rankers like BM25.

Building on these ideas, several learned sparse retrieval models have been developed. Early models like DeepCT [39] and SPARTA [218] focused on learning term weights. Similarly, DeepImpact [108] learns "semantic impact scores" for terms using contextualized language models, leading to enhanced performance on metrics such as MRR@10, NDCG@10, and MAP. More recent models have refined this approach. uniCOIL [56] introduces a simplified scalar term weighting mechanism, while SPLADE v2 [51] uses BERT to create sparse document representations that include term-level importance scores, showing strong results on benchmarks like MS MARCO and BEIR. Furthermore, SpaDE [31] incorporates both term weighting and expansion through a dual encoder architecture, with its effectiveness demonstrated across multiple datasets using metrics such as MRR@10, Recall@K, and NDCG@10.

A distinct category of sparse methods focuses on learning sparse vector representations directly. UHD-BERT [81], for instance, generates ultra-high dimensional sparse vectors. Perhaps the most radically different design is the Standalone Neural Ranking Model (SNRM) [211], which learns sparse latent representations for both queries and documents, allowing them to be searched directly via an inverted index without a separate first-stage retrieval model.

3.4 Hybrid and Multi-Vector Approaches

To harness the complementary strengths of different architectures, hybrid and multi-vector approaches integrate sparse and dense signals or employ multi-stage ranking pipelines. For example, the Duet Model combines a local model for exact lexical matches with a distributed model for semantic matching. CEDR (Contextualized Embeddings for Document Ranking) [120] integrates contextualized word representations like BERT into traditional neural ranking architectures. A common multi-stage pipeline might use monoBERT (pointwise re-ranker) and duoBERT (pairwise re-ranker) after an initial retrieval stage to progressively refine results, balancing effectiveness and latency [137].

3.5 Context-Aware and Personalized Retrieval

Recent advances have introduced models that capture user behavior, conversational structure, and document intent for improved relevance. COCA (Contrastive learning for Context-Aware document ranking) [224] models user behavior sequences using contrastive learning. For conversational IR, Hierarchical Dependence-aware Aggregation (HDA) [47] models conversations as graphs to capture contextual dependencies. In multi-turn response selection, Sequential Matching Network (SMN) [193] and Multi-hop Selector Network (MSN) [208] match utterances chronologically, with MSN using a multi-hop strategy to prioritize relevant context. For personalized search, some models integrate short- and long-term user behavior. Additionally, Graph4DIV [171] explicitly models document intents through a dynamic graph structure to enhance result diversity. Collectively, these models highlight the growing emphasis on context, user behavior, and intent in developing more intelligent retrieval systems.

Metric Name	Category	Description
Precision	Traditional Retrieval Quality	Ratio of relevant retrieved documents to the total retrieved documents.
Precision@k	Traditional Retrieval Quality	Proportion of relevant documents within the top k results.
Recall	Traditional Retrieval Quality / Completeness	Proportion of all relevant items that are successfully retrieved.
Average Precision	Metrics Averaged Over Queries	Mean precision at ranks where relevant documents are retrieved.
Mean Average Precision (MAP)	Metrics Averaged Over Queries	Average of AP scores across all queries for overall effectiveness.
Inferred Average Precision (infAP)	Metrics Averaged Over Queries (Incomplete Judgments)	Estimates AP with incomplete judgments by assuming unjudged documents mirror judged ones.
Reciprocal Rank (RR)	Metrics Averaged Over Queries	Reciprocal of the rank of the first relevant document.
Expected Reciprocal Rank (ERR)	Gain-Based / User Behavior	Simulates user browsing with graded relevance.
Top-k Accuracy	Traditional Retrieval Quality / Completeness	Frequency that a relevant document appears in the top k results.
Coverage	Completeness of Retrieval	Proportion of all relevant documents appearing in the top k results.
Robustness	Robustness & Reliability	Evaluates retrieval quality under domain shift, noise, or ambiguity.
Efficiency	System-Oriented Efficiency	Assesses speed, scalability, and resource usage of retrieval systems.
Bpref	Metrics Averaged Over Queries (Incomplete Judgments)	Considers only judged documents, penalizing relevant ones ranked below judged non-relevant ones.
Rejection Rate	Robustness & Reliability	Measures how often the system correctly declines when no relevant information is available.
Error Detection Rate	Robustness & Reliability	Assesses the system's ability to detect and filter incorrect or misleading information.
Faithfulness	Robustness & Reliability	Checks generated text accurately reflects retrieved documents.
Discounted Cumulative Gain (DCG)	Gain-Based	Scores relevance by weighting higher-ranked documents more heavily using a logarithmic discount.
Alpha-DCG and Alpha-nDCG	Gain-Based / Diversity	Extend DCG/NDCG by penalizing redundant documents for multi-intent queries.
Normalized Cumulative Gain (NCG)	Gain-Based	Computes cumulative relevance without position discounting, unlike NDCG.
Single-Query Latency (ms)	System-Oriented Efficiency	Measures total processing time for a single query in milliseconds.
Batch Throughput	System-Oriented Efficiency	Number of queries the system can handle per second in parallel.
Memory Footprint	System-Oriented Efficiency	Total memory required for model parameters, indexes, and caches.
Knowledge F1 (K-F1)	Knowledge Coverage	Token-level F1 overlap between generated output and ground-truth knowledge.
Unique-Facts Coverage	Diversity & Redundancy	Measures how many distinct relevant facts are retrieved without repetition.
Redundancy Penalty	Diversity & Redundancy	Penalizes repeated facts by comparing unique facts to total occurrences.
Misleading Rate (MR)	Robustness & Reliability	Frequency with which the system produces misleading or incorrect answers.
Mistake Reappearance Rate	Robustness & Reliability	Tracks how often previously made errors reappear in system outputs.

Table 1. Evaluation Metrics, Category and Description for Retrieval

The evaluation system leverages standard IR metrics including NDCG@10 and MRR@100, tested on the widely used MS MARCO document retrieval dataset. Experimental results show that DANCE significantly improves retrieval performance over previous state-of-the-art dense retrieval methods (e.g., ANCE).

This paper investigates how Large Language Models (LLMs) can act as knowledge generators in place of traditional information retrieval (IR) systems for knowledge-intensive tasks such as open-domain question answering and knowledge-grounded dialogue.

4 Retriever Evaluation

Evaluating the retriever component of a RAG system is a multifaceted process that extends beyond traditional information retrieval. It involves not only assessing the quality of ranked results but also the utility of the retrieved information for the downstream generator, the overall behavior of the system, and the methodologies used to ensure fair and reproducible assessment. Therefore, retriever-centric evaluation focuses on isolating this component, employing metrics and frameworks that assess its performance independent of the final language model's output. This section provides a comprehensive overview of retriever evaluation, organized into four key areas: core ranking quality metrics, information-centric evaluation, system-level and downstream impact, and the frameworks that enable modern evaluation (Table 1).

4.1 Core Ranking Quality Metrics:

These metrics evaluate the fundamental task of a retriever that rank relevant documents as highly as possible. They are typically calculated for individual queries and then averaged across a test set to produce a single performance score.

4.1.1 ***Ranking Metrics for Binary Relevance.*** A primary evaluation approach is Accuracy which measure ranking quality, which quantifies a system's ability to prioritize relevant documents at the top of its results. This is typically done by comparing the system's output against a ground truth of human relevance judgments.

Among the most fundamental metrics for this is Precision [20] is defined as the ratio of relevant retrieved documents to the total number of retrieved documents. A key variant, Precision at k (P@k) [13], measures the proportion of relevant documents within the top *k* retrieved results, emphasizing early precision. This metric quantifies exactness by answering: "Of the top k results, how many are truly relevant?" This aligns with its broader interpretation in information retrieval: a measure of result quality, where higher precision indicates fewer irrelevant documents in the retrieved set. For example, if the top 5 results for a query contain 3 relevant documents, the P@5 would be 3/5 = 0.6. The formula for Precision@k is:

$$\text{Precision@k} = \frac{\text{Number of questions with an answer in the top k documents}}{\text{Total number of questions}}$$

However, precision alone can be misleading, as a system could achieve a perfect score by returning only one correct document. To measure the completeness of the results, Recall [125] is used as a complementary metric. It measures the proportion of all relevant items that are found in the top k results. It is crucial for RAG systems, as a failure to retrieve a relevant document in the first stage means the information is lost forever. High recall ensures the generator has access to all needed information. It's a measure of completeness, answering the question: "Of all the possible relevant items, how many did the system retrieve in the top k?" The formula for Recall@k is:

$$\text{Recall@k} = \frac{\text{Number of relevant items in the top k results}}{\text{Total number of relevant items}}$$

For instance, if there are a total of 10 relevant documents for a query and the top 5 results contain 4 of them, the Recall@5 would be 4/10 = 0.4. Common variations include Recall@50 and Recall@1000 for reflecting the system's ability to retrieve relevant documents at various cut-off points effectively [153].

The weighted harmonic mean of precision and recall F1-score which is useful when both retrieval completeness and concentration of relevant documents matter.

$$\mathrm{F} = \frac{2*\mathrm{Precision}*\mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

Mean Average Precision (MAP) provides a comprehensive assessment of a ranking system's effectiveness by averaging the Average Precision (AP) scores across a set of queries. For a single query, AP measures ranking quality by averaging the precision value at the position of each retrieved relevant document. This mechanism heavily rewards systems that place relevant documents early and consistently, making MAP highly sensitive to the rank of every relevant item [21, 224]. A common variation, MAP@k, considers only the top k results, which is useful in scenarios where users are unlikely to examine the full list.

The formula for Average Precision for a single query is:

$$\mathrm{AP} = \frac{1}{N_{\mathrm{rel}}} \sum_{k=1}^{n} P(k)\,\mathrm{rel}(k)$$

n is the total number of documents retrieved. $P(k)$ is the precision at rank k, and $rel(k)$ is an indicator function that is 1 if the item at rank k is relevant and 0 otherwise. MAP is then the mean of these AP scores across all queries.

While MAP evaluates the ranking of all relevant documents, Mean Reciprocal Rank (MRR) focuses on a simpler, often crucial, task: how effectively a system ranks the first relevant document. For each query, the "reciprocal rank" is the inverse of the rank of the first correct answer ($1/rank_i$). MRR is then the average of these scores across all queries. This makes MRR ideal for tasks like question answering where finding the first correct result is the primary goal. However, its main limitation is that it does not account for any relevant documents beyond the first one. A common variation, MRR@k, considers only the top k results. If the first relevant item is not found within the top k, the reciprocal rank for that query is 0. This is useful in applications where users are unlikely to look beyond a certain number of results, with common cutoffs being MRR@10 and MRR@100 [103].

$$\mathrm{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\mathrm{rank}_i} \tag{1}$$

Here, $|Q|$ is the total number of queries and $rank_i$ is the rank of the first relevant results for the i-th query. It is frequently reported at a cutoff, such as MRR@10 or MRR@100 [103].

An even simpler metric, often used in similar QA contexts, is Top-k Accuracy. It measures how often the correct passage is found within the top k retrieved results (e.g., k=20 or 100). For a single query with one correct answer, the score is 1 if the answer is in the top k results and 0 otherwise. When averaged over many queries, this is equivalent to Recall@k for single-relevance tasks and was a key metric used in the evaluation of Dense Passage Retrieval (DPR) [87].

4.1.2 ***Ranking Metrics for Graded Relevance.*** Traditional IR evaluation methods, based on binary relevance judgments (relevant vs irrelevant), do not differentiate between documents that are marginally or highly relevant. This work [82] introduced graded relevance judgments, where documents can have different degrees of relevance.

Normalized Discounted Cumulative Gain (NDCG) is a sophisticated metric that measures ranking quality by incorporating two key principles: graded relevance and positional importance. The process begins with Discounted

Cumulative Gain (DCG), which calculates a score for a ranked list by rewarding documents with high relevance scores and penalizing them for appearing lower in the results. It does this by applying a logarithmic discount to each document's relevance value based on its rank, reflecting the real-world user behavior of valuing top results more. The normalized version, NDCG, makes the metric comparable across different queries by scaling the DCG score with the score of an ideal, perfect ranking (IDCG). This creates a final score between 0.0 and 1.0 and provides a nuanced assessment of ranking quality that accounts for varying degrees of document usefulness.

$$\text{CG@k} = \sum_{i=1}^{k} \text{rel}_i$$

where rel_i is the graded relevance of the result at position i. Then $CG@k$ with a logarithmic discount for items at lower ranks to measure the discounted cumulative gain (DCG@k).

$$\text{DCG@k} = \sum_{i=1}^{k} \frac{\text{rel}_i}{\log_2(i+1)}$$

At last steps, the DCG@k score normalized by the Ideal DCG (IDCG@k), which is the DCG score of a perfect ranking to make Normalized Discounted Cumulative Gain (NDCG@k). This normalization ensures the metric's value is between 0 and 1.

$$\text{NDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}}$$

It is considered a primary metric in many benchmarks, with common cutoffs at k=1, 3, 5, 10, and 20 [175, 224]. To overcome the limitations of binary (relevant/irrelevant) evaluation, a graded relevance judgment was introduced [82].

To evaluate qualities beyond simple relevance, such as novelty and diversity, an extension of NDCG called alpha-nDCG was developed. It is an extension of the NDCG metric, proposed by Clarke et al. [34], designed specifically to evaluate novelty and diversity in search results. It works by adding a penalty for redundant documents in the ranked list, which is particularly useful for queries that have multiple intents or subtopics. By rewarding new information and penalizing repetition, alpha-nDCG provides a more nuanced score for search quality beyond simple relevance. This metric belongs to a broader family of measures focused on diversity, which also includes Expected Reciprocal Rank for diversified IR (ERR-IA), Normalized Rank Biased Precision (NRBP) [131], and Subtopic Recall (S-Rec).

Another powerful metric that utilizes graded relevance is the Expected Reciprocal Rank (ERR) [25], which models user behavior directly. The ERR metric models realistic user behavior by simulating how searchers interact with ranked results. It assumes users examine documents sequentially and stop once they find a result that satisfies their information need. Unlike binary metrics, ERR incorporates graded relevance probabilities to account for varying degrees of document usefulness. This approach captures two key aspects of real-world search behavior: (1) user persistence - the likelihood that a searcher will continue examining lower-ranked results if unsatisfied, and (2) graded relevance - the recognition that documents provide different levels of usefulness. By weighting each position in the ranking according to the probability that it will satisfy the user, ERR provides a more nuanced evaluation of retrieval effectiveness that better aligns with actual search experiences.

4.2 Information-Centric Evaluation:

These metrics assess the intrinsic qualities of the retrieved content itself, focusing on whether it is complete, current, and factually sufficient for the task.

4.2.1 ***Coverage and Completeness:*** Coverage, often used interchangeably with Recall@k, is a crucial metric that measures the proportion of all known relevant documents successfully retrieved within the top-k results. High coverage is vital for downstream tasks like Question Answering (QA), as the answer-generation component depends entirely on the retriever to supply the necessary facts. In a QA context, Coverage@k is often defined as the percentage of questions for which at least one of the top-k retrieved documents contains the correct answer. For example, open-domain QA and fact-checking benchmarks like FEVER [176] and HotpotQA [205] evaluate evidence retrieval based on the proportion of "gold" evidence pieces found. Some frameworks also define a more explicit comprehensiveness score, calculated as the fraction of unique relevant facts from the ground truth that appear in the retrieved results [55]. Ultimately, a high coverage score indicates that the system is not missing important information required for the downstream task.

4.2.2 ***Knowledge Coverage:*** These metrics evaluate how well the retrieved information aligns with the specific knowledge required to answer a query. Fact-level or Knowledge Coverage metrics evaluate how well the retrieved information aligns with the specific knowledge required to answer a query. One such metric is Knowledge F1 (K-F1), which measures the token-level F1 overlap between the retrieved passages and a ground-truth knowledge snippet [2]. By computing the harmonic mean of precision and recall based on the overlap of words, it assesses both the accuracy and completeness of the retrieved information at a granular level, indicating that the retriever has found passages containing the necessary facts.

While K-F1 provides a balanced measure, Coverage Rate focuses specifically on the completeness of the retrieved knowledge. Acting as a form of "knowledge recall," it measures the fraction of ground-truth facts, entities, or information nuggets that are present in the retrieved passages [206]. This provides a more detailed assessment than document-level recall; for instance, a sub-question coverage rate can check how many key components of a complex question are addressed by the retrieved chunks. Ultimately, these fact-level metrics help quantify whether the retriever has successfully found not just the right documents, but the specific, essential pieces of knowledge within them.

4.2.3 ***Timeliness of Information:*** Timeliness evaluates whether a retriever provides up-to-date information, a crucial factor for queries about recent events or in domains where knowledge evolves rapidly (e.g., science or technology). This is a key advantage of RAG systems, as a retriever linked to a live knowledge source can supply current facts that an LLM's frozen parametric knowledge lacks [104]. While there isn't a single, standardized "timeliness score," it is typically measured in one of two ways: 1. Timestamp Verification: Checking the publication or modification dates of the retrieved documents. 2. Fact Verification: Confirming that the retrieved content includes the latest known information, which is common in news retrieval or dynamic QA benchmarks. Some research also incorporates freshness and recency as direct ranking factors, ensuring that newer documents are prioritized for time-sensitive queries. In practice, evaluating timeliness involves rewarding retrievers that consistently return the most current information relevant to a query while avoiding stale or outdated sources.

4.3 System-Oriented Evaluation

Beyond assessing the ranking of relevant documents, a comprehensive evaluation of a RAG retriever must consider its practical, system-level attributes. This holistic view, known as system-oriented evaluation, examines the quality of the entire result set, the system's stability under challenging conditions, and its computational performance, which are all critical for real-world deployment.

4.3.1 ***Diversity and Redundancy.*** For complex or ambiguous queries, it is crucial that a retriever provides a diverse set of results rather than repetitive information. This quality is measured using specialized metrics that reward novelty and penalize redundancy. A family of such metrics includes alpha-nDCG, which extends traditional NDCG by penalizing documents that cover already-seen aspects of a topic, as well as Expected Reciprocal Rank for diversified IR (ERR-IA), Normalized Rank Biased Precision (NRBP), and Subtopic Recall (S-Rec) [15, 34]. The underlying principle of these metrics can also be measured more directly as Unique-Facts Coverage, which quantifies how many distinct pieces of information from the ground truth are present in the retrieved set. The inverse of this is the Redundancy Penalty, often calculated as a ratio that decreases as the same facts are repeated, thereby rewarding outputs that are concise and information-dense.

4.3.2 ***Robustness and Reliability.*** A retriever's robustness is its ability to maintain high-quality performance under non-ideal conditions, such as zero-shot domain shifts or when faced with noisy, ambiguous, or adversarial queries. Beyond general robustness, a system's reliability is often measured by its capacity to avoid generating misinformation. The Misleading Rate gauges how often a model produces a confident but incorrect answer, especially when the necessary information is absent from its knowledge base. Mistake-Reappearance Rate tracks error consistency, measuring how frequently a model repeats a mistake it has already made. Finally, the Error-Detection Rate evaluates a system's self-awareness, or its ability to recognize and flag its own potential errors by expressing uncertainty instead of stating a falsehood.

4.3.3 ***Efficiency and Scalability.*** For a retriever to be viable in a real-world application, it must be not only accurate but also fast, efficient, and scalable. Evaluation of these characteristics focuses on three key areas. First, speed is measured by Single-Query Latency (often as Mean Response Time), which is the end-to-end time to process one query, and Time to First Token (TTFT), which is crucial for interactive applications. Second, scalability is measured by Batch Throughput, which gauges how many queries the system can handle in parallel per second. Third, cost is measured by Memory Footprint, which includes the size of the model, indexes, and caches. These metrics are heavily influenced by the retrieval strategy; a baseline like Brute-Force Dense Search may offer high recall but suffers from poor latency and cannot scale, whereas systems using Approximate Nearest Neighbor (ANN) indexing trade a small amount of accuracy for massive improvements in speed and throughput [55, 123].

4.4 Evaluation Frameworks and Methodologies:

Evaluating a RAG retriever requires more than just calculating scores; it depends on the underlying experimental paradigms, tools, and theoretical principles that ensure the evaluation is both meaningful and practical. This includes classic test collection-based approaches but focuses on modern solutions like LLM judges and automated frameworks such as ARES [158], which are designed to handle the scale and cost challenges inherent in evaluating large-scale RAG systems. Furthermore, this section covers principled techniques for handling real-world data imperfections, such as incomplete relevance judgments, and explores the theoretical concepts that validate the entire evaluation process.

4.4.1 ***Test Collection-Based Evaluation:*** The classic offline evaluation method for information retrieval is the Cranfield paradigm [35]. This approach relies on a static test collection composed of three key parts: a fixed set of documents, a set of queries, and a list of human relevance judgments. To create the pool of documents that require manual judging, modern techniques like query-variant pooling are used. This method builds test collections at a lower cost by automatically generating query variants and using multiple retrieval models (e.g., BM25, TF-IDF) to

gather a diverse set of candidate documents, reducing the reliance on many participating systems [138]. By running different retrieval systems against the final collection, researchers can conduct rigorous, reproducible experiments. While foundational, the Cranfield paradigm has significant limitations for evaluating modern RAG systems. First, its assumption of a static nature, where relevance is fixed and independent of user intent, fails to capture the dynamic, conversational, and context-dependent queries that RAG systems often handle. Additionally, the sheer cost and scale of RAG applications make the Cranfield approach impractical; creating comprehensive relevance judgments for vast and constantly evolving knowledge bases is often prohibitively expensive, leading to incomplete and potentially biased assessments.

4.4.2 ***LLM as a Judge:*** The "LLM as a Judge" [220] approach has emerged as a scalable and nuanced alternative to manual annotation for retriever evaluation. In this paradigm, a powerful language model (e.g., GPT-4) is prompted to act as an expert, assigning a relevance score (typically on a 1-5 scale) to each document retrieved for a given query. This method excels at capturing complex semantic relevance beyond simple keyword matching and can be applied to large test sets where human annotation would be prohibitively expensive; for instance, the RAGAS [44] framework demonstrated that GPT-3.5 can reliably rate context relevance. The trustworthiness of these automated judges is established by measuring their agreement with human experts. This is done by reporting correlation coefficients (such as Pearson's r or Spearman's ρ) between the LLM's scores and a held-out set of human relevance labels. A strong correlation indicates that the LLM judge is a reliable and trustworthy proxy for human evaluation.

4.4.3 ***Frameworks for Automated Evaluation:*** Modern RAG retriever evaluation increasingly relies on automated frameworks that address the challenges of scale, cost, and complexity. A prime example is ARES [158], which first employs Synthetic Data Generation to create question-answer pairs from an unlabeled corpus, training a lightweight "judge" model to assess context relevance without expensive human annotation. It then uses Prediction-Powered Inference (PPI) to calibrate this judge with a small, human-annotated validation set, providing statistically rigorous metrics with minimal manual effort. Beyond static relevance, these methodologies also assess dynamic retrieval capabilities, such as the effectiveness of Context Selection Mechanisms in conversational RAG [208] and Dynamic Query Reformulation in systems that must adaptively select and combine information sources [6].

4.4.4 ***Handling Incomplete Judgments:*** Evaluating RAG retrievers over vast knowledge bases presents a significant challenge, as creating complete relevance judgments is often impractical. This makes traditional metrics like Mean Average Precision (MAP) unreliable, since they introduce bias by treating all unjudged documents as irrelevant. To address this, specialized metrics have been developed to provide reliable scores from sparse labels. Two prominent approaches address this issue with different strategies. Often used as a more stable alternative to MAP and Precision when relevance judgments are incomplete, Bpref (Binary Preference) works by ignoring unjudged items entirely. It focuses only on judged documents and, for each relevant item, calculates the proportion of judged non-relevant items ranked above it [226]. In contrast, the second approach, Inferred Average Precision (infAP), treats unjudged documents probabilistically, assuming they contain relevant items in the same proportion as the judged set and scaling this ratio to provide a more reliable AP approximation. Both methods offer more robust and stable scores than MAP in the sparsely-labeled environments common to RAG systems.

4.4.5 ***Advanced Theoretical Concepts:*** This area of research delves into the theoretical underpinnings of IR evaluation, examining both the validity of traditional metrics and exploring novel ways to create more predictive ones. Defending the classic approach, Moffat et al. [130] address critiques against metrics like NDCG and MAP, arguing

that they remain meaningful and valid despite using non-interval scales. In a complementary line of work, Chuklin et al. [32] explore a novel methodology, showing that evaluation metrics derived from user click models correlate more strongly with live online A/B-test outcomes than traditional offline metrics, particularly when relevance judgments are incomplete.

Metric	Type	Description
ROUGE-1,2,L [107]	N-gram-based	Measures overlap of n-grams, subsequences, and skip-bigrams between system outputs and human references.
ROUGE-C	Quality-centric	Evaluates informativeness and readability using the source document instead of gold references.
BLEU [140]	N-gram-based	Computes n-gram overlap with penalty to discourage overly short outputs.
BERTScore [215]	Embedding-based	Aligns tokens with contextual embeddings to assess semantic similarity.
METEOR [14]	N-gram-based	Uses stemming, synonyms, and weighted recall–precision harmonic mean for paraphrase-sensitive evaluation.
BARTScore [209]	Advanced	Uses BART to compute text likelihood, assessing informativeness, fluency.
MoverScore [219]	Embedding-based	Measures minimum embedding "travel distance" between outputs and references with soft alignments.
BLEURT [163]	Embedding-based	Fine-tunes Transformers on human and synthetic data to score grammar, semantics, and fluency.
COMET [151]	Embedding-based	Uses source, candidate, and reference with human-labeled training to assess adequacy and fluency.
Cosine Sim. [152]	Similarity-based	Compares vector embeddings of texts via cosine angle for meaning similarity.
Jaccard similarity	Similarity-based	Measures token set overlap as intersection over union.
SPICE [5]	Advanced	Converts text into scene graphs to evaluate object, attribute, and relation correctness.
BLANC [180]	Quality-centric	Assesses summary informativeness by measuring reduction in MLM perplexity.
SUPERT [60]	Quality-centric	Builds pseudo-references from source sentences and compares semantic overlap with summaries.
FactCC [94]	Entailment-centric	Uses a weakly supervised classifier to detect factual consistency or contradictions.
SummaC [96]	Entailment-centric	Applies NLI-based entailment aggregation to identify inconsistencies in summaries.
SRLScore [48]	Factuality / QA-centric	Compares predicate–argument structures via Semantic Role Labeling to assess factual consistency.
QuestEval [162]	Factuality / QA-centric	Frames evaluation as QA, checking answerability precision and recall for summaries.
QAFactEval [46]	Factuality / QA-centric	Enhances QA-based evaluation by decomposing question generation, answering, and overlap.
Grammaticality	Linguistic Quality Metric	Checks if the generated text follows grammatical rules and is syntactically correct.
Readability	Linguistic Quality Metric	Measures how easy the text is for a human reader to understand.
Coherence	Linguistic Quality Metric	Evaluates the logical flow and connectedness of ideas across sentences and paragraphs.
Fluency	Linguistic Quality Metric	Assesses the naturalness and smoothness of text, making it sound human-written.

Table 2. Various metrics for RAG generator alongside type and description.

5 Generator Evaluation

Evaluating the quality of the LLM's generated output, given relevant documents from the retriever, remains a challenging task—particularly for open-ended or creative responses. Traditional evaluation compares generated text against ground truth references across multiple dimensions such as linguistic quality, coherence, accuracy, and topical alignment. Surface-level metrics like BLEU and ROUGE-L measure n-gram overlap, providing insights into how well the generated content captures key phrases and ideas. Exact Match (EM) and F1 assess accuracy and overlap with reference answers, with EM requiring fully correct responses and F1 capturing partial matches. Fluency and diversity are often approximated using Perplexity and Entropy, which evaluate naturalness, grammaticality, and response variability, particularly in dialogue settings. To measure robustness, task-specific metrics such as the Misleading Rate and Mistake Reappearance Rate quantify an LLM's tendency to repeat errors or generate factually incorrect statements. For tasks with multiple valid answers, metrics such as Micro-F1 [158] and Prediction Accuracy [79] provide finer-grained assessments of correctness (Table 2).

5.1 Classical Metrics

Classical metrics have long served as the foundation for automatic generator evaluation, generally falling into two categories. The first includes reference-based methods like the n-gram overlap scores ROUGE [107] and BLEU [140], and the more semantically-aware embedding-based metrics like BERTScore [215]. While these metrics are inexpensive and easy to compute, they share significant limitations for evaluating complex RAG outputs: they often correlate weakly with human judgments, depend on costly "gold" references that may be unavailable, and fail to capture semantic accuracy or factual correctness. Another category includes intrinsic measures like Perplexity, which assesses a model's fluency by measuring how well it predicts a sequence of text.

5.1.1 ***N-gram Metrics:*** N-gram–based metrics have long been the cornerstone of automatic text generation evaluation, providing simple and inexpensive proxies for quality by measuring the surface-level overlap between a system's output and human-written references. The most common examples include BLEU [140], which measures n-gram precision with a brevity penalty, and ROUGE [107], a recall-oriented metric that has become the standard for summarization. More advanced variations like METEOR [14] improve upon these by incorporating stemming and synonym matching, while TER (Translation Edit Rate) [166] takes a different approach by calculating the number of edits required to transform the output into the reference. However, a critical limitation unites all these metrics: they are fundamentally restricted to surface-level similarity and often fail to capture deeper semantic meaning or factual correctness, which can result in weak correlations with human judgments in complex domains [135].

5.1.2 ***Embedding metrics****:* Embedding-based metrics were developed to overcome the limitations of n-gram methods by using contextualized embeddings to capture semantic similarity, making them more robust to paraphrasing. Approaches like BERTScore [215] compute a score by aligning tokens in the generated and reference texts based on their embedding similarity, while MoverScore [219] calculates the minimum "travel distance" between the two sets of embeddings using an optimal transport method. More advanced metrics are explicitly trained for the evaluation task; BLEURT [163] is a regression model fine-tuned on human ratings to predict a quality score, while COMET [151] also incorporates the original source text to evaluate both adequacy and fluency. Although these methods generally show stronger correlation with human judgments than n-gram metrics, studies have found this alignment can be inconsistent in complex domains, raising concerns about their generalizability [135].

5.1.3 ***Similarity-Based Metrics****:* Similarity-based metrics offer lightweight and interpretable ways to assess how close a generated response is to a reference or its source context. One common approach, Cosine Similarity [152], operates in a vector space by converting texts into embeddings and measuring the angle between them to gauge overall semantic closeness; a technique sometimes used in RAG to compare the generated answer to the retrieved context. In contrast, Jaccard Similarity is a token-set-based metric that measures lexical overlap by calculating the intersection over the union of unique words in the two texts. While both are intuitive, they generally provide only a coarse-grained judgment and lack the diagnostic power required for a nuanced evaluation of a RAG generator's output.

5.1.4 ***Advanced metrics:*** Beyond n-gram and embedding-based measures, a class of advanced metrics has been developed to capture deeper aspects of generation quality, such as informativeness and factuality, by integrating pretrained models or task-specific structures. A key example is BARTScore [209], which reframes evaluation as a text-generation task. As RAG systems expand into multimodal domains, metrics from image captioning are also becoming increasingly relevant. CIDEr [181] models human consensus by weighting salient concepts against multiple human references, while SPICE [5] enables a more fine-grained evaluation by converting captions into scene graphs to assess the correctness of semantic propositions about objects, attributes, and relationships.

5.2 Faithfulness

To address the failure of traditional metrics in assessing factual correctness, a class of faithfulness and factual consistency metrics has been developed to evaluate whether a generated response is consistent with its source evidence. These approaches are often reference-free, grounding the evaluation in the retrieved context, and include quality-oriented methods like 'SUPERT' [60] and 'BLANC' [180] that measure informativeness, as well as entailment-based metrics like 'FactCC' [94] and 'SummaC' [96] that use Natural Language Inference (NLI) to detect contradictions or unsupported claims. It's important to distinguish between faithfulness, which requires the generated answer to strictly adhere to the retrieved context, and factual consistency, which demands correctness against a trusted external ground truth. Despite their effectiveness, these methods face practical challenges for large-scale RAG evaluation, as they often require running large, costly models for each claim and depend heavily on annotated training data.

5.2.1 ***Quality-Centric Metrics****:* Quality-centric metrics evaluate the informativeness and usefulness of a generated response without relying on human-written references, making them highly scalable for RAG evaluation. One approach, SUPERT [60], creates a "pseudo-reference" by extracting the most salient sentences from the retrieved documents and then measures the semantic overlap with the generated text using contextual embeddings. In contrast, BLANC [180] takes a different approach by measuring how much the generated summary improves the performance of a Masked Language Model (MLM), using the reduction in perplexity as a proxy for informativeness. Together, these lightweight, reference-free methods offer a practical way to assess the utility and quality of a generator's output, often correlating more strongly with human judgments than traditional n-gram metrics.

5.2.2 ***Entailment-Centric Metrics:*** Entailment centric metrics are crucial for evaluating the factual faithfulness of a RAG generator by verifying whether the generated text is logically supported by the retrieved documents. Unlike surface-level similarity measures, these approaches directly test for consistency, making them highly effective at detecting hallucinations, contradictions, or entity swaps. Prominent examples include FactCC [94], which uses a BERT-based classifier trained on synthetically altered sentences to predict consistency, and SummaC [96], which adapts Natural Language Inference (NLI) to resolve the granularity mismatch between single sentences and the full document

context. Together, these entailment-based methods provide a scalable and systematic way to ensure that a generator's output is factually grounded in the retrieved evidence, which is essential for any task where reliability is critical.

5.2.3 ***Factuality / QA-Centric Metrics:*** Factuality and QA-centric metrics address a key failure of traditional metrics by interrogating generated text for truthfulness, which is vital for detecting hallucinations in RAG systems. These approaches move beyond surface similarity by using structured facts or question answering to verify claims. One method involves decomposing text into structured semantic units, such as the predicate-argument tuples in SRLScore [48] or the atomic facts in FACTScore [128], which are then verified against a knowledge source. Another popular paradigm frames evaluation as a Question-Answering task such as QuestEval [162] generates questions from the output to test against the source (precision) and vice-versa (recall), a method that QAFactEval [46] systematically improves. Other methods measure grounding more directly. The Attribution Score calculates the proportion of generated tokens traceable to retrieved passages, while FFLM [83] offers a zero-shot faithfulness metric that measures probability shifts to detect hallucinations without relying on large proprietary models.

Recent benchmarks and methodologies have further refined factuality evaluation and introduced robustness testing. Frameworks like TrueTeacher [61] train smaller, efficient factuality evaluators on large-scale LLM-annotated data, while FHSumBench [144] specifically targets the detection of "factual hallucinations"—claims that are factually correct but unsupported by the retrieved context, a critical distinction for RAG. Rather than relying on external sources, SelfCheckGPT [124] provides a lightweight approach by checking for inconsistencies across multiple responses generated from the same query. To test for robustness, Contrastive Evaluation introduces adversarial distractors into the retrieved context to see if the generator blindly trusts misleading information, a stress test conceptually related to the TruthfulQA [109] benchmark. These issues can be quantified with a straightforward Factual Error Rate, which measures the proportion of claims in an answer that are either unsupported by or contradicted by the retrieved documents. Together, these approaches form a comprehensive toolkit that moves RAG generator evaluation beyond surface similarity toward genuine factual reliability.

5.2.4 ***Hallucination Detection & Severity.*** Hallucination Detection is a critical aspect of RAG generator evaluation that assesses whether a generated answer is properly supported by the retrieved documents, with faithfulness metrics designed to penalize unsupported content. For a RAG system, this evaluation highlights whether generated claims are fully grounded in the retrieved evidence. However, simply detecting a hallucination is often not enough; differentiating minor slips from critical factual errors provides more actionable insights into the generator's reliability. To formalize this, a Hallucination Severity Index can be used, which recognizes that not all hallucinations are equally harmful and assigns different weights based on their real-world impact. In this index, minor hallucinations (e.g., slight imprecision) incur a small penalty; major hallucinations (clearly incorrect but not safety-critical) receive a medium penalty; and critical hallucinations (e.g., providing an incorrect medical dosage) carry the highest penalty due to their potential for serious harm. The final index is computed as a weighted sum across these categories, making it especially valuable for evaluating RAG systems in high-stakes domains like healthcare, law, and finance, where balancing fluency with factual accuracy is essential.

5.3 LLM as a Judge

Large Language Models (LLMs) such as GPT-4 [1], Claude, and Llama [178] are increasingly used as judges to evaluate generative outputs. Unlike traditional metrics, which rely on n-gram overlap or embedding similarity, LLM-based evaluation can flexibly assess multiple quality dimensions such as, fluency, coherence, consistency, relevance, and

factual grounding, without requiring human-written references. A key motivation is human alignment, where model judgments are validated against human scores or preferences. LLM judges also improve scalability by offering more reproducible and cost-effective evaluations than human annotators, though they must address challenges such as reliability across prompts, positional bias, and susceptibility to over-trusting LLM-generated text. For RAG systems, LLM judges are especially useful: they can verify whether generated claims are supported by retrieved documents, assess contextual relevance, and highlight hallucinations that reference-based metrics may miss.

5.3.1 ***Scalar Scoring.*** One of the most common formats for LLM-based evaluation is scalar scoring, where models assign numeric ratings to generated outputs. These typically follow Likert-style scales (e.g., 1–5 or 0–100) and are applied across diverse quality dimensions, including fluency, coherence, consistency, relevance, grammaticality, and informativeness. Frameworks like LLM-EVAL [110] unify these multiple aspects into structured evaluations that do not require gold references. While this approach offers simplicity and flexibility, the scores can be sensitive to prompt design and model biases. To improve interpretability, many systems enhance this method by pairing scalar scores with short textual justifications, allowing the LLM judge to explain the reasoning behind its ratings. In the context of RAG evaluation, scalar scoring is particularly useful for producing multi-dimensional "trust profiles" of generated responses. An LLM judge can assign separate numerical ratings for key RAG qualities: fluency (linguistic quality), relevance (alignment with the query), and faithfulness (consistency with retrieved passages). For faithfulness specifically, this can be adapted to classify each claim as “Supported,” “Partially Supported,” or “Unsupported” based on the retrieved evidence. These fine-grained numeric judgments make it possible to track trade-offs between language quality and factual grounding, enabling more consistent and detailed benchmarking of RAG systems across different domains.

5.3.2 ***Prompt-Based Scoring.*** Prompt-based scoring enhances the stability and reliability of LLM evaluations by structuring prompts into explicit rubrics, checklists, or reasoning templates rather than just asking for an opaque scalar score. Foundational methods include CheckEval [98], which improves inter-evaluator agreement using binary yes/no checklists; the Reason-Then-Score (RTS) paradigm, which increases transparency by requiring the LLM to justify its judgment before scoring; and head-to-head preference selection, which frames the task as a Multiple-Choice Question (MCQ) to pick the better of two outputs. While these structured methods typically correlate better with human judgments than automatic metrics like ROUGE, they can still suffer from inconsistency and bias [164].

More advanced techniques extend this paradigm to zero-shot setups and more complex reasoning. GPTScore [54], for example, leverages zero-shot instruction prompts to directly generate quality scores, demonstrating that LLMs can rate summaries on dimensions like clarity and accuracy without gold references. Taking this a step further, Error Analysis Prompting (EAPrompt) [118] combines Chain-of-Thought reasoning with explicit error categorization to mimic the human Multidimensional Quality Metrics (MQM) framework. By requiring the LLM to identify major and minor errors before scoring, EAPrompt produces evaluations that are more explainable and human-like; however, its applicability depends on the underlying LLM’s capability and carries a risk of test data leakage.

In practice, prompt-based evaluation can be applied in two complementary modes: (i) evaluating generated text in isolation to assess intrinsic qualities like fluency and coherence, and (ii) evaluating generated text alongside external context to judge faithfulness. For RAG generator evaluation, the second mode is especially useful. Structured prompts can explicitly require the LLM judge to assess whether claims are grounded in the retrieved passages, making faithfulness judgments more interpretable and providing a targeted method for detecting hallucinations.

5.3.3 ***Chain-of-Thought Grading.*** Chain-of-Thought (CoT) grading builds on prompt-based approaches to improve both the reliability and interpretability of LLM-based evaluation. Instead of producing direct scalar ratings, this process requires the LLM evaluator to first generate a structured rationale explaining its judgments, which is then translated into a fine-grained score, reducing unexplained variance and increasing transparency. A leading example is G-EVAL [115], which prompts GPT-4 for a reasoning chain before scoring dimensions like coherence and relevance, and uses token-probability normalization to generate continuous ratings. While G-EVAL significantly outperforms earlier evaluators on major benchmarks, analysis also reveals a bias towards favoring LLM-generated text over human-written text. Extending this line of reasoning, Error Analysis Prompting (EAPrompt) [118] combines CoT with explicit error categorization inspired by the human MQM framework, guiding the LLM to identify major and minor errors before scoring to produce more interpretable and human-like judgments, succeeding even at the challenging segment level.

Beyond general-purpose evaluation, specialized CoT judges have been developed for specific stages of the model lifecycle. PandaLM [189], for instance, is a judge LLM designed specifically for evaluating outputs during instruction tuning. Unlike benchmarks that focus on correctness, PandaLM assesses subjective qualities such as conciseness, clarity, and instruction-following. Trained on human-annotated data, the open-source PandaLM models perform on par with or even surpass proprietary models like GPT-4, while crucially avoiding costly API calls and reducing data leakage risks, making them reproducible and privacy-preserving tools for model development.

Together, these CoT-based methods demonstrate that explicitly reasoning through evaluation criteria yields more reliable, interpretable, and human-aligned judgments than direct scoring. For RAG generator evaluation, CoT grading is particularly valuable. It allows an LLM judge to reason step-by-step about whether each claim in a generated answer is properly supported by the retrieved evidence, offering a fine-grained and trustworthy assessment of a RAG system's grounding and factual faithfulness.

5.3.4 ***Role-Playing.*** While single-agent "LLM-as-a-judge" methods, particularly those using Chain-of-Thought, improve upon traditional metrics, they can still fall short of human-level reliability. To bridge this gap, recent work, inspired by multi-annotator human practices, explores multi-agent and role-playing frameworks. A leading example is ChatEval [24], which introduces a debate-style setup where multiple LLM agents assume different roles and engage in structured discussions before reaching a collective judgment, leading to more transparent and human-like assessments. Similarly, DRPE (Diverse Role-Player Evaluation) [191] leverages LLMs as both objective role-players (e.g., grammar judges) and subjective ones (e.g., readers with varied perspectives), aggregating votes to capture nuanced dimensions like coherence and interestingness that other methods often miss. Experiments show these collaborative approaches achieve higher consistency with human annotators than both traditional metrics and direct single-LLM scoring.

For RAG generator evaluation, these role-playing methods are especially valuable because different evaluators can be assigned specific roles such as "fact checkers," "domain experts," or "general readers." This multi-perspective setup ensures that generated answers are scrutinized not only for fluency and coherence but also for their grounding and faithfulness to the retrieved evidence. By simulating a team of human reviewers, this approach reduces the risk of overlooking subtle errors or hallucinations, making the overall RAG evaluation process more robust and trustworthy.

5.3.5 ***Pairwise and Comparative Evaluation.*** As an alternative to assigning absolute scores, pairwise and comparative evaluation methods align closely with human judgment by simplifying the task to a direct comparison between two candidate outputs. A prominent example is Pairwise Ranking Prompting (PRP) [145], which improves robustness over pointwise or listwise prompting and has shown that moderate-sized open-source models can achieve state-of-the-art ranking performance. This core idea has been extended in several ways: PAIRS [116] systematically uses pairwise

judgments to generate full rankings, similar to RLHF; Goal-Reversed Prompting (GRP) [168] improves robustness by asking the LLM to identify the worse output rather than the better one; and the Elo Rating system from chess has been adapted to create dynamic, leaderboard-style benchmarks from repeated pairwise comparisons.

Studies on Comparative Assessment [117] confirm that pairwise evaluation is often more effective and intuitive than scalar scoring, especially for mid-sized open-source LLMs that struggle with direct absolute scoring. However, this line of research also reveals that these methods can suffer from systematic positional biases (e.g., favoring a candidate in the first position), though simple debiasing strategies can mitigate this issue. For RAG generator evaluation, these comparative frameworks are especially valuable. They provide a robust and interpretable method for ranking multiple candidate responses generated from the same retrieved evidence, making them critical for the key RAG task of selecting the single best-grounded and most reliable answer.

5.3.6 ***Human with GPT-Based Evaluation***. While automated evaluation methods like Chain-of-Thought and pairwise comparison offer scalability, hybrid evaluation methods aim to achieve human-level reliability by combining human oversight with GPT-based scoring. The core principle is that humans provide trustworthy ground-truth checks and handle ambiguous edge cases, while LLM judges ensure efficiency, consistency, and reproducibility across large datasets. A direct application of this is ChatGPT-assisted annotation, where an LLM acts as a tool to help human evaluators by flagging potential factual errors, detecting hallucinations, or pre-ranking system responses, thereby reducing manual annotation cost and effort.

More structured hybrid frameworks have also been developed. As discussed previously, systems like G-Eval [115] and PandaLM [189] exemplify this by using LLMs to apply detailed rubrics that are often derived from or validated by human-annotated data. A more specialized system, FENCE [195], introduces a fine-grained factuality evaluator trained on augmented datasets where each claim is paired with critiques and external evidence (from search engines or knowledge graphs). By providing such detailed claim-level feedback, FENCE enables an LLM to revise or remove its own false claims, demonstrably improving the factuality of models like Llama3-8B by over 14% on the FActScore benchmark.

A recent evolution of this paradigm involves using ensembles of LLMs as judges. For instance, Badshah et al. [11] propose a reference-guided approach where multiple diverse LLMs issue binary True/False judgments on a candidate output, which improves reliability and reduces prompt sensitivity. The WideDeep framework [216] extends this further by arranging LLMs into evaluator networks with specialized roles (e.g., correctness, coherence, relevance) and aggregating their judgments, akin to an academic peer-review process. For RAG generator evaluation, this blend of human and machine evaluation is particularly effective. It provides a robust methodology for verifying that generated answers are not only fluent but remain factually grounded in the retrieved evidence, thereby minimizing hallucinations in high-stakes applications.

5.3.7 ***Alternative Prompting and Reasoning Strategies***. Beyond the structured prompting methods like Chain-of-Thought and pairwise comparison, a new wave of alternative prompting and reasoning strategies aims to further mitigate common weaknesses of LLM evaluators, such as bias and inconsistency. One such approach is Goal-Reversed Prompting (GRP) [168], which inverts the traditional pairwise paradigm by prompting the LLM to choose the worse of two responses. This simple shift, which mirrors human critical reasoning, has been shown to reduce positional and preference biases and consistently improve evaluation accuracy on benchmarks like JudgeBench [198] across top-tier models. Another promising method is EvalPlanner [160], which separates evaluation into two distinct stages: first, it trains an LLM to generate a step-by-step evaluation plan tailored to the task, and only then executes that plan to

deliver scores. By disentangling planning from reasoning, EvalPlanner achieves state-of-the-art, data-efficient results on benchmarks such as RewardBench and JudgeBench, leading to more transparent and robust judgments than those from rigid, pre-defined rubrics.

Together, these alternative strategies are particularly valuable for RAG generator evaluation. By encouraging more systematic reasoning, they can reduce hallucination bias and improve the quality of grounding checks. For RAG systems, the key benefit is that these methods ensure the LLM evaluator focuses not only on surface similarity but also on the underlying reasoning and grounding processes that determine whether a generated answer is truly supported by the retrieved evidence.

5.3.8 ***Advanced Systems.*** Moving beyond the specific prompting techniques discussed earlier, a class of Advanced Systems has emerged to overcome the persistent challenges of bias, inconsistency, and lack of interpretability in "LLM-as-a-Judge" frameworks. These systems extend beyond simple scoring by incorporating structured reasoning, modular pipelines, and specialized training to achieve higher robustness and alignment with human judgment. For RAG generator evaluation, such systems are particularly valuable because they can verify factual grounding, detect hallucinations, and provide interpretable feedback at scale. This line of work can be broadly categorized by its focus: enhancing the reasoning and explainability of the judge, creating modular and cost-effective architectures, or training highly specialized evaluators.

One major focus is on improving the reasoning process and explainability of the evaluation itself. SOCREVAL [68], for example, evaluates reasoning ability using Socratic-style questioning, while REPS (Rationale Enhancement through Pairwise Selection) [90] improves reliability by iteratively refining rationales in a tournament-style setup to filter out flawed explanations. A leading example of an explainable framework is TIGERScore [84], a universal evaluator built on LLaMA-2 that provides instruction-guided error analysis and assigns penalty scores for specific error types. While it achieves state-of-the-art correlation with human judgments, challenges remain with its potential to hallucinate errors and its slower inference speed.

Another direction focuses on building more modular, cost-effective, and specialized judge systems. VERDICT [86] takes a compositional approach by orchestrating modular reasoning units, such as verification, debate, and aggregation into compound judge systems to improve robustness and serve as runtime guardrails. Addressing the high cost of evaluation, RocketEval [190] uses a powerful model to generate instance-specific checklists, which are then graded by a much smaller, lightweight model, cutting evaluation costs by over 50 times while achieving performance comparable to GPT-4o. This trend toward specialization is also seen in frameworks like AUTO-J [102] and Self-Taught Evaluators [187], which train specialized judge models that outperform general-purpose LLMs. Other systems like LLM-EVAL [110] aim to streamline the process with a single-prompt schema to score multiple dimensions in one call.

These advanced judge systems are being extended into diverse domains like code generation and patch validation, where they can outperform human crowdworkers in consistency and efficiency [29, 62, 69]. However, this entire line of research underscores the importance of reliable validation practices to ensure the judge systems themselves are trustworthy, especially under ambiguous conditions [66]. Across all these advanced frameworks, persistent challenges remain, including susceptibility to biases (positional, verbosity, self-enhancement), instability across reruns, and the ethical concerns of replacing human oversight. In summary, advanced judge systems like TIGERScore, VERDICT, and RocketEval represent the next generation of LLM-based evaluation: they are interpretable, modular, and scalable. For RAG generator evaluation, they are especially powerful because they can go beyond surface similarity to directly assess factual grounding, reasoning validity, and reliability—key factors for deploying trustworthy RAG systems.

5.4 Meta-Evaluation

Having established various evaluation metrics and "LLM-as-a-Judge" frameworks, meta-evaluation shifts the focus to assessing the evaluators themselves—measuring how well they align with human judgments and remain reliable under diverse conditions. For RAG generator evaluation, this is especially critical, as a weak or biased evaluator may misjudge faithfulness or factual grounding, leading to misleading system comparisons. While traditional approaches rely on simple correlation with human ratings (using Spearman's ρ or Kendall's τ), these methods can be undermined by inconsistent human annotations and averaging practices that obscure genuine differences. By systematically analyzing correlation, robustness, and fairness, meta-evaluation frameworks ensure that the metrics we use for RAG tasks are trustworthy and scalable.

To address the issues of simple correlation, more sophisticated meta-evaluation frameworks have been developed. Some impose greater structure on the process such as, Dual-Perspective Evaluation [74] introduces global (ordinal classification) and local (pairwise comparison) views to capture both coarse and nuanced reliability, while the Branch-Solve-Merge (BSM) pipeline [159] decomposes evaluation into dimensions to reduce biases and improve human alignment. Other methods focus on the evaluator's output, such as distributional judging, which leverages the full probability distribution of scores for better calibration [188]. Further efforts tackle bias and reliability directly where some learn a linear mapping between LLM judgments and human ratings to help smaller models approximate larger ones [179], while benchmarks like LLMBAR [212] stress-test evaluators with adversarial cases, revealing that even models like GPT-4 fall short of human reliability in such settings.

Finally, research has highlighted that the quality of data used to train evaluators is a critical concern. The D3 framework [214] addresses this by developing criteria—diversity, difficulty, and dependability—to select smaller, higher-quality datasets for instruction-tuning judge models, enabling them to achieve superior performance with a fraction of the training data. In sum, meta-evaluation research shows that no single method is sufficient; distributional reasoning improves calibration, structured frameworks capture nuanced performance, and alignment strategies mitigate bias. For RAG generators, these methods are essential to ensure that the automated evaluators we rely on can faithfully assess grounding in retrieved evidence, which is critical for avoiding the overestimation of system performance in knowledge-intensive tasks.

5.5 Fine-Tuned and Explainable Evaluators

While zero-shot "LLM-as-a-Judge" systems provide strong general-purpose evaluation, a new class of fine-tuned and explainable evaluators aims for superior alignment with human judgments and greater transparency. These systems are trained specifically for the evaluation task, often on datasets of human preferences or model-generated critiques. A leading example is JudgeLM [223], which fine-tunes open-source LLMs on GPT-4-generated judgments and incorporates specific techniques like swap augmentation to mitigate common positional and format biases. Similarly, InstructScore [200] trains a LLaMA model on synthetic data to provide not only a score but also interpretable, natural-language explanations for its evaluation, demonstrating stronger performance than zero-shot GPT models on benchmarks like COMET [151].

Other research focuses on improving the fine-tuning process and promoting open-source solutions. TRACT [30] enhances reliability by combining Chain-of-Thought (CoT) reasoning with a regression-aware loss in a two-stage training process, more effectively coupling the model's reasoning quality with its final score calibration. In the pursuit of reproducibility, OpeNLGauge [88] introduces a fully open-source, explainable metric built on open-weight LLMs.

Available as both an ensemble of large models and a distilled fine-tuned version, it provides fine-grained, error-span-based explanations, offering a transparent and efficient alternative to proprietary evaluators.

The scope of fine-tuned evaluators is also expanding to more adaptive and multimodal domains. ARJudge [199] adaptively generates task-specific criteria and integrates both text-based and code-driven analyses, allowing it to assess structural and quantitative constraints that other evaluators overlook. This specialization is also seen in frameworks like CritiqueVLM [194], which are fine-tuned for specific contexts like Chinese multimodal tasks, achieving higher human alignment than general-purpose models like GPT-4. Together, these fine-tuned and explainable evaluators demonstrate that domain specialization, error-traceability, and adaptive strategies significantly enhance performance. For RAG generator evaluation, they are particularly valuable as they can provide interpretable, claim-level feedback and better bias control, making them highly suited for tasks where factual support and evidence attribution are critical.

5.6 Collaborative Evaluation

While single-agent "LLM-as-a-Judge" systems offer scalability, Collaborative Evaluation frameworks aim for even greater reliability, fairness, and interpretability by combining multiple evaluators—either LLMs, humans, or both. Inspired by peer review and debate, these approaches seek to overcome the weaknesses of single-judge systems, such as bias and prompt sensitivity. For RAG generator evaluation, such methods are particularly useful because they allow multiple perspectives (e.g., consistency with retrieved context, factual correctness, and user preference) to be debated or reconciled, leading to more trustworthy and nuanced assessments.

One line of work focuses on enabling multiple LLM agents to collaborate with each other. The multi-agent debate framework ChatEval [24] has LLMs take on diverse roles and engage in structured discussions before reaching consensus. A more structured approach, GraphEval [50], decomposes complex ideas into graph nodes and models their relationships to achieve fairer outcomes and even detect novelty. Other methods focus on improving inter-evaluator agreement and efficiency. CheckEval [98] replaces subjective Likert scoring with binary checklists to increase consistency, while the DAFE [12] arbitration framework improves efficiency by invoking a third "arbitrator" judge only when the first two disagree.

Another major direction involves creating hybrid systems that combine human and machine intelligence. This includes frameworks like G-Eval [115] and PandaLM [189] and the practice of ChatGPT-assisted annotation, where LLMs augment human evaluators. Yue et al. [210] demonstrates how GPT-4 can be used for large-scale attribution evaluation, categorizing references in a human-like way. A key finding in this area is that while human evaluation is costly (around 1 USD per query [113]), using human-written references significantly improves agreement and reduces the self-preference bias of LLM judges [93]. This collaboration can also be structured into feedback loops, where verbal critiques are fed back into generation prompts. An advanced version of this is Chain-of-Hindsight (CoH) [111], which conditions models on both positive and negative feedback to outperform standard fine-tuning.

Finally, some frameworks create a deeply interactive, human-in-the-loop process. Kim et al. [91] developed a system where users iteratively refine prompts with LLM-based feedback, balancing automation with user-defined criteria. Despite these advanced automated and hybrid methods, the need for reliable human-grounded verification remains critical. A human audit of generative search engines by Liu et al. [113] found that only 51.4% of generated sentences were properly supported by their citations. This underscores the value of collaborative approaches—whether through multi-agent debate, arbitration, or human-LLM hybrids—for RAG generator evaluation, as they provide a more robust and interpretable path to verifying that generated content is both semantically relevant and faithfully grounded in retrieved evidence.

5.7 Specialized Applications

While the previously discussed frameworks—from fine-tuned models to collaborative evaluators—provide powerful general-purpose tools, their true value is often demonstrated in Specialized Applications where they are adapted to domain-specific settings. This is particularly relevant for RAG systems, where outputs often require not only fluency but also domain-sensitive measures like citation quality or factual grounding. For example, in Automated Essay Scoring (AES), LLM evaluators have been adapted using zero-shot, few-shot, and p-tuning strategies, which use trainable prefix vectors to align the model more closely with the scoring task [170]. In a different context, the ALICE benchmark for Citation Quality evaluates whether a generator's claims are adequately supported by its citations, measuring citation precision in addition to correctness [57]. Although this is constrained by the accuracy of underlying NLI models, it is critical for RAG systems where ensuring traceable and faithful grounding is paramount.

Another key application focuses on improving the reliability of the judge itself, especially in user-centric contexts. Research into Confidence-Aware and Personalized Judging has shown that LLM judge reliability decreases when persona information is sparse. To address this, methods that incorporate verbal uncertainty estimates have been proposed, allowing the model to down-weight its own low-confidence judgments, which significantly improves consistency with human ground truth [41]. For RAG evaluation, such confidence-aware judging is highly valuable, as it helps filter unreliable grounding assessments and improves the overall factual reliability of the system. Ultimately, these specialized applications illustrate how LLM-based evaluators can be tailored for domain-specific needs, allowing RAG pipelines to enforce not just standard qualities like fluency, but also the critical dimensions of grounding, traceability, and trustworthiness in their generated outputs.

6 End-to-End RAG Evaluation

For a holistic understanding, the evaluation of a Retrieval-Augmented Generation (RAG) system must move beyond assessing the retriever and generator in isolation. Instead, it requires an end-to-end approach that assesses not only the quality of the final output but also the performance and reliability of the entire pipeline. This section explores the key pillars of this assessment, from holistic automated frameworks and system performance metrics to tests for robustness and the critical role of human-centric evaluation (Table 3).

6.1 Holistic Automated Frameworks

While the previous sections detailed metrics for the retriever and generator in isolation, a true end-to-end RAG evaluation must assess the tight coupling between them, since errors in either module can propagate and distort final outputs. This requires a holistic view built upon five key metrics: Answer Relevance and Faithfulness for the generator, and Context Relevance, Context Recall, and Context Precision for the retriever. Together, these complementary metrics provide a comprehensive view of the entire pipeline's performance, allowing for targeted diagnosis of failures.

A widely adopted framework that formalizes these metrics is RAGAS [44]. It is reference-free and leverages LLMs to evaluate faithfulness, answer relevance, and context relevance. For faithfulness, candidate answers are decomposed into atomic statements, which are then verified against retrieved passages via structured yes/no judgments. Answer relevance is measured by generating sub-questions from the answer and comparing their embeddings to the original question's embedding. Context relevance is assessed by having the LLM extract the minimal subset of sentences needed to answer the query from the context. Validated on the WikiEval benchmark, RAGAS demonstrates high alignment with human judgment and integrates seamlessly with toolkits like LangChain. Building on this, ARES [158] addresses some

Metric Name	Model	Description
Answer Relevance	RAGAS [44]	Measures if the answer directly addresses the question by comparing original and generated question embeddings.
Faithfulness	RAGAS	Ensures answers are grounded in the provided context by verifying decomposed statements against it.
Context Relevance	RAGAS	Assesses if retrieved context contains only necessary information by comparing crucial extracted sentences.
Context Recall	RAGAS	Checks whether the retrieved context includes all information required to answer the question.
Contextual Precision	RAGAS	Evaluates if relevant chunks appear near the top of retrieval, formalized as mean precision@k.
Answer Relevance	ARES [158]	Evaluated using a fine-tuned, lightweight judge model.
Faithfulness	ARES	Assessed by a specialized, fine-tuned judge model to verify that the answer is grounded in the provided context.
Context Relevance	ARES	Assesses passage pertinence using a fine-tuned classifier calibrated with PPI.
Noise Vulnerability	MIRAGE [196]	Measures how easily a model gives wrong answers when presented with noisy or mixed context.
Context Acceptability	MIRAGE	Evaluates whether the model can use context effectively to generate accurate answers despite noise.
Context Insensitivity	MIRAGE	Identifies cases where the model ignores context and produces incorrect answers regardless of input.
Recall (Coverage)	RAGCHECKER [157]	Measures how comprehensively retrieved claims capture all relevant information.
Context Precision (Relevance)	RAGCHECKER	Measures how relevant retrieved claims are, helping balance recall and noise.
Faithfulness	RAGCHECKER	Evaluates the percentage of generated claims supported by retrieved context.
Noise Sensitivity	RAGCHECKER	Measures incorrect claims caused by either relevant/irrelevant chunks.
Context Utilization	RAGCHECKER	Measures the proportion of retrieved relevant claims actually used in the generated response.
Semantic Evaluation (LLMeval)	BERGEN [150]	An LLM judge semantically evaluates answer correctness to enable standardized, reproducible experiments.
Downstream-Derived Relevance	eRAG [161]	Leverages the system's own LLM to score each document's utility for the final task, yielding a highly correlated relevance score.
Full-Chain Analysis	CoFE-RAG [112]	Evaluates all stages of the RAG pipeline, from chunking and retrieval to generation, for a comprehensive, full-pipeline analysis.
Systematic Component Comparison	RAG Playground [139]	An open-source tool for systematic comparison of different RAG configurations, such as various retrieval or prompting strategies.
Automated Dataset Generation	RAGEval [222]	A framework for automatically generating scenario-specific evaluation datasets and metrics to address the high cost of data creation.
Long-Context vs. RAG Comparison	U-NIAH [59]	A unified framework for comparing RAG systems against long-context LLMs in "Needle-in-a-Haystack" scenarios.
Black-Box Deployed Evaluation	SCARF [154]	Enables black-box evaluation of fully deployed RAG systems to assess end-to-end performance in a real-world setting.
Vulnerability to Data Poisoning	Poison-RAG [134]	Assesses a system's vulnerability to adversarial data poisoning attacks, particularly in RAG-based recommender systems.

Table 3. End-to-End RAG Evaluation Metrics.

of RAGAS's limitations by using synthetic training data to fine-tune lightweight judge models (e.g., DeBERTa-v3-Large) for each metric. To ensure robustness, ARES then applies prediction-powered inference (PPI), which calibrates these automatic judgments using a small set of 150–300 human-labeled datapoints to produce statistical confidence intervals.

While RAGAS and ARES focus on faithfulness and relevance, subsequent frameworks introduced deeper diagnostic capabilities. MIRAGE [141] expands evaluation to capture a system's adaptability under noisy retrieval conditions by introducing four mutually exhaustive metrics: Noise Vulnerability, Context Acceptability, Context Insensitivity, and Context Misinterpretation. This fine-grained decomposition allows MIRAGE to diagnose why a system fails; for example, a high Context Insensitivity score indicates a generator integration failure rather than a retriever error. Complementing this, the Retrieval-Augmented Generation Benchmark (RGB) was introduced specifically to test these fundamental abilities, revealing that current models struggle significantly with Noise Robustness, Negative Rejection, Information Integration, and Counterfactual Robustness. To further tighten the retrieval-generation alignment, RAGCHECKER [157] introduces claim-level entailment checking. It decomposes answers into individual claims and evaluates them using a three-tiered structure of overall, retriever, and generator diagnostics. This diagnostic depth provides actionable insights for system design and correlates more strongly with human judgment than previous frameworks.

Other frameworks address the practical challenges of conducting rigorous and efficient end-to-end evaluation. To improve experimental rigor, the BERGEN [150] framework provides an open-source library for standardized and reproducible experiments, establishing best practices such as the need for strong retrieval with a reranking stage. Addressing the high computational cost and poor correlation of some methods, the eRAG framework [161] offers a novel solution by using the RAG system's own LLM to evaluate each retrieved document individually. This approach is not only more efficient but also produces retriever scores that have a significantly higher correlation with true end-to-end task performance.

Several novel evaluation frameworks have been introduced to provide more comprehensive, full-chain analysis and to assess specific system qualities. For full-pipeline analysis, CoFE-RAG [112] evaluates all stages from chunking to generation, while RAG Playground [139] offers an open-source tool for systematic comparison of different RAG configurations. To address the high cost of data creation, RAGEval [222] provides a framework for automatically generating scenario-specific evaluation datasets and metrics. Other frameworks target specific capabilities, such as the evaluation of fairness across demographic attributes [192], the comparison of RAG and long-context Large Language Models with U-NIAH [59], the black-box evaluation of deployed systems with SCARF [154], and the assessment of a system's vulnerability to adversarial data poisoning attacks with Poison-RAG [134].

The underlying technique for many of these systems is the general "LLM-as-a-Judge" approach, where large models like GPT-4 are prompted to directly score outputs for relevance, faithfulness, and completeness, offering a scalable and reference-free alternative. In summary, end-to-end RAG evaluation has evolved from simple overlap metrics toward these multi-faceted, claim-level, and adaptability-aware frameworks. RAGAS pioneered reference-free scoring, ARES improved scalability with fine-tuned judges, MIRAGE and the RGB benchmark captured robustness to noise, and RAGCHECKER introduced claim-level diagnostics. Together, these methods chart a path toward holistic, reliable, and actionable evaluation of RAG pipelines.

6.2 System Performance and Efficiency

While the holistic frameworks discussed previously assess the accuracy of a Retrieval-Augmented Generation system, its practical viability in real-world applications also depends on its system performance and efficiency. Beyond correctness, a system must deliver timely responses, handle high query loads, and remain resource-efficient to support deployment at

scale. The most direct measure of user experience is Latency, which refers to the total time a RAG system takes to process a query and produce a final answer, spanning all stages from query processing and embedding generation to document retrieval and Large Language Model inference. Typically reported in milliseconds or seconds, low latency is vital for interactive applications like chatbots where users expect near-instantaneous responses. This overall measurement can be decomposed into retrieval and generation times to identify bottlenecks, guiding whether to optimize indexing algorithms or apply techniques like model compression. Importantly, while reducing latency improves responsiveness, aggressive optimization must always be balanced against the risk of compromising response quality.

Beyond the speed of a single query, a system's scalability is evaluated by its ability to handle high query loads and remain resource-efficient. The primary metric for this is Throughput, which measures the number of queries a system can process per second under a steady workload, reflecting its ability to handle multiple user requests efficiently over time. This is closely related to Parallel-Query Efficiency, which evaluates how throughput is maintained under concurrent loads; a system that shows minimal performance degradation when multiple queries are handled simultaneously demonstrates strong scalability. Another critical factor for scalability is the Peak Memory Footprint, which captures the maximum Central Processing Unit or Graphics Processing Unit memory consumed during query processing. This includes the storage of document indexes, inference-time context, and generation buffers. Systems with lower memory footprints can handle more parallel queries on the same hardware, which directly reduces deployment costs and improves operational efficiency in production environments.

6.3 Robustness and Reliability

In addition to being efficient, a truly production-ready Retrieval-Augmented Generation system must be robust and reliable, maintaining high-quality outputs even when faced with imperfect, noisy, or adversarial conditions. This evaluation focuses on the system's ability to handle misleading inputs without degrading answer quality. A key indicator of this is Information Diversity; a systems that repeatedly surface similar or redundant content risk overlooking important evidence, so measuring diversity ensures that the retrieved and generated content covers a broader range of relevant facts. This extends to Noise Robustness, which is the system's ability to handle irrelevant or misleading content within the retrieved context. This is evaluated using metrics such as the Misleading Rate—how often irrelevant information is used—and the Mistake Reappearance Rate—how often previously seen errors are repeated—to ensure the system learns to ignore distractors and prioritize verified information.

A further dimension of robustness involves the system's response to insufficient or deliberately false information. Negative Rejection is the critical capability of a system to abstain from answering when it lacks sufficient evidence. Instead of fabricating answers, a robust system will explicitly indicate that a question cannot be reliably answered from the retrieved content, a crucial safeguard in high-stakes domains like healthcare or law. The ultimate test of this is Counterfactual Robustness, which assesses the system's ability to resist plausible but false information. This is tested by introducing adversarial distractors that mimic relevance but contain incorrect facts. A system that incorporates these distractors into its output demonstrates an over-reliance on retrieval without sufficient reasoning, whereas a counterfactually robust system will filter such content to maintain factual accuracy and preserve its trustworthiness even under adversarial conditions.

6.4 Human-in-the-Loop and User-Centric Evaluation

While automated metrics for performance and robustness provide scalable assessments, human-in-the-loop and user-centric evaluation remain indispensable for validating Retrieval-Augmented Generation systems, especially in high-stakes or domain-specific scenarios. Human involvement is critical for capturing nuanced insights that automated systems may miss. This is often implemented through hybrid scores, which combine automated metrics like BLEU, ROUGE, or BERTScore with human ratings collected via Likert scales or pairwise comparisons. This human-centric evaluation also emphasizes diverse perspectives, making a key distinction between expert evaluation, which offers higher accuracy for specialized tasks but is slower and costlier, and crowd-sourced evaluation, which provides a cheaper and faster alternative. For sensitive domains like healthcare or finance, expert review loops are particularly important, as domain specialists are needed to validate outputs for factual correctness and subtle errors that non-experts would overlook.

To make this process more effective, evaluators often use tools and scalable protocols that bridge automated metrics with human oversight. Human-in-the-loop dashboards provide assessors with a unified interface that displays the query, generated answer, retrieved evidence, and automated scores for faithfulness or hallucination rates. By consolidating these elements, dashboards allow for rapid verification, in-depth error analysis, and transparent feedback collection. To scale this efficiently, protocols such as batched evaluation (processing multiple queries simultaneously) and distilled local LLM judges—smaller models fine-tuned to approximate stronger evaluators—are employed to offer faster and more cost-effective assessments for large-scale monitoring. Even when general GPT-based scoring is used, it should be complemented with human validation to ensure fairness and reliability.

Beyond static assessment, user-centric evaluation can be integrated directly into the model improvement process through reinforcement learning. Approaches such as Reinforcement Learning from Human Feedback (RLHF) and Reinforcement Learning from AI Feedback (RLAIF) create iterative evaluation loops. In these systems, the model first generates an output, which is then critiqued by a human or another AI system. The model is subsequently updated using this feedback to better align with the desired style, factuality, or quality. Over time, these iterative feedback loops progressively refine the model's ability to produce context-grounded, coherent, and trustworthy outputs, directly connecting the evaluation process to model development.

7 Benchmark Datasets

7.1 Retriever-Focused Corpora

Information retrieval (IR) research relies heavily on standardized corpora for evaluation. Over the decades, these corpora have evolved from small, domain-specific collections to large-scale, heterogeneous benchmarks that capture the diversity and complexity of real-world queries. Classic resources such as the Cranfield collections, TREC Ad Hoc tracks, and Reuters–RCV1 established foundational methodologies, while more recent datasets such as MS MARCO, BEIR, and MIRAGE reflect the needs of modern neural retrieval and retrieval-augmented generation (RAG) systems (Table 4, 5).

7.1.1 ***Foundational Test Collections:*** Early work was shaped by the Cranfield collections (Cranfield-1 & 2), which introduced exhaustive relevance judgments and established the Cranfield evaluation methodology [35]. Cranfield-1 included 1,398 aeronautical abstracts and 225 queries, while Cranfield-2 expanded to 1,300 documents and 221 queries. These pioneering datasets laid the groundwork for systematic test collection design in IR.

Dataset	Type	One-Sentence Description
TREC Ad Hoc [38]	General English IR	Large-scale collections with graded relevance judgments for ad hoc retrieval tasks, widely used as classic IR benchmarks.
MS MARCO [136]	Open-domain QA / Passage Retrieval	Over 1M Bing queries with passages and human-written answers, used for passage ranking, QA, and document retrieval.
NTCIR	Cross-lingual / Multilingual IR	Collections for multilingual and cross-lingual retrieval tasks across Asian and English languages.
CLEF	Cross-lingual / Multilingual IR	Evaluation campaigns with multilingual test collections supporting IR, QA, and medical retrieval tasks.
InPars [16]	Synthetic IR Dataset	Uses LLMs to generate question–document pairs filtered by likelihood to fine-tune smaller retrievers like monoT5.
Medical IR (PubMed) [17]	Domain-Specific Biomedical IR	Thousands of layman-English health queries linked to PubMed abstracts with graded relevance labels.
BEIR [174]	Heterogeneous / Zero-shot IR Benchmark	Unifies 18 datasets across 9 task types, enabling zero-shot evaluation of lexical, dense, sparse, and hybrid models.
Cranfield (1 & 2) [35]	Classic IR Benchmark	Early aeronautical abstract collections (1,300 docs, 220 queries) introducing exhaustive relevance judgments and the Cranfield paradigm.
Reuters RCV1 [99]	Text Categorization / News IR	800K newswire stories labeled with hierarchical categories across topics, industries, and regions.
20 Newsgroups [4]	Topic Classification / Document Retrieval	20K Usenet posts across 20 topics (politics, science, computers, sports), widely used for classification and topic modeling.
FIRE [129]	Multilingual / Legal / Domain-Specific IR	Annual benchmark for Indian languages (Hindi, Bengali, etc.) and specialized tasks such as legal retrieval.
MIRAGE [141]	Adversarial / RAG Robustness	37.8K passages crafted with positive/negative samples to evaluate robustness of retrieval-augmented generation systems.
MTEB [132]	Massive Embedding Benchmark	50+ datasets across retrieval, classification, and bitext mining for benchmarking embedding models.
Natural Questions (NQ) [95]	Open-domain QA / Retriever	307K real Google queries paired with full Wikipedia pages, annotated with short and long answers.
HotpotQA [205]	Multi-hop QA Retrieval	113K Wikipedia-based QA pairs requiring multi-hop reasoning with supporting facts for explainability.
TriviaQA [85]	Complex QA Retrieval	650K question–answer–evidence triples from trivia questions and web/Wikipedia sources, requiring multi-sentence reasoning.
WikiQA [204]	Open-domain QA Retrieval	3K Bing search queries linked to Wikipedia, annotated at sentence level with many unanswerable queries.

Table 4. Benchmark Datasets for Retriever Evaluation.

Dataset	Type	One-Sentence Description
PopQA [122]	Entity-centric QA Retrieval	14K Wikidata-based entity questions emphasizing long-tail knowledge beyond popular entities.
IfQA [207]	Counterfactual QA Retrieval	3.8K counterfactual "if"-based QA pairs requiring reasoning beyond factual recall.
KILT [143]	Unified Knowledge-Intensive Benchmark	5.9M Wikipedia articles consolidating 11 datasets across tasks (fact checking, QA, entity linking) with provenance spans.
IFIR [169]	Instruction-Following Retrieval	2,426 examples across finance, law, healthcare, and science with multi-level instruction-following queries.
Specialized Collections [37, 71, 167]	Domain-Specific IR	Includes MEDLINE/MEDLARS & TREC Genomics (biomedical), TREC Legal (jurisprudence), and TREC Microblog (social media).

Table 5. Benchmark Datasets for Retriever Evaluation.

In parallel, Reuters–RCV1 [99] and 20 Newsgroups [4] became standard benchmarks for text classification and categorization. RCV1 is a large, manually categorized test collection containing over 800,000 Reuters newswire stories, labeled with a hierarchical category system covering topics, industries, and regions. It serves as a benchmark resource for developing and evaluating classification and related NLP methods.

The 20 Newsgroups dataset, by contrast, includes about 20,000 Usenet posts evenly distributed across 20 topics such as politics, religion, science, sports, and computers. Organized into broader categories, it is widely used for evaluating algorithms in document classification, topic modeling, and text mining. It has a moderate size, diverse, and relatively clean structure dataset.

7.1.2 ***General-Domain IR Benchmarks.*** Building on this foundation, the TREC Ad Hoc tracks greatly expanded both scale and scope. A notable example is the TREC 2022 Deep Learning Track, which built on the MS MARCO v2 passage and document collections. MS MARCO v2 contains 11.9 million documents from which 138 million passages were derived, and the final evaluation set includes 76 topics for both passage and document ranking tasks [38]. This track prioritized creating a more reusable and complete test collection, with document relevance inferred from passage-level judgments, marking a significant step forward in large-scale evaluation.

Beyond retrieval alone, many corpora directly support open-domain question answering (QA) and have become central to evaluating RAG systems. MS MARCO [136], built from real Bing search queries with over 1 million human-written answers, has become a cornerstone for machine reading comprehension, passage ranking, and generative QA. Unlike synthetic QA datasets, it emphasizes real-world, diverse, and often ambiguous queries, making it particularly valuable for training and evaluating both retrievers and generators. It also underpins the TREC Deep Learning tracks, serving as the foundation for large-scale passage and document ranking benchmarks.

Complementing MS MARCO, the Natural Questions (NQ) dataset [95] uses anonymized Google search queries paired with full Wikipedia pages. Annotators provide both long answers, such as paragraphs, tables, or lists, and short answers, such as entities or yes/no, with robust 5-way annotations for evaluation. Similarly, HotpotQA [205] introduces 113,000 Wikipedia-based QA pairs designed for multi-hop reasoning and explainability, requiring models to integrate

information from multiple documents. It includes diverse question types and provides sentence-level supporting facts, enabling strong supervision.

Further increasing reasoning complexity, TriviaQA [85] contains over 650K question–answer–evidence triples, with 95K human-authored questions and evidence gathered from Wikipedia and the web. The dataset features naturally occurring, compositional, and diverse questions that demand multi-sentence reasoning and integration of world knowledge. Its difficulty is underscored by the large performance gap between baseline models (40% EM) and human accuracy (80%). Finally, WIKIQA [204] provides over 3,000 Bing queries linked to Wikipedia pages, focusing on answer sentence selection and triggering. Many questions have no correct answers, making it useful for evaluating systems on both answer detection and relevance.

Together, these QA-focused datasets—MS MARCO, NQ, HotpotQA, TriviaQA, and WIKIQA—progressively raise the challenge from single-hop factual retrieval to multi-hop reasoning, answer triggering, and explainable QA, making them essential benchmarks for advancing both retrievers and generators in RAG systems.

7.1.3 ***Specialized / Domain-Specific Corpora.*** In the medical domain, Boteva et al. [17] introduced a full-text learning-to-rank dataset by linking thousands of layman-English health queries from PubMed to research article abstracts. This dataset includes queries of varying lengths and styles with three levels of graded relevance—direct, indirect, and topic-based—enabling the training of complete learning-to-rank models rather than relying on preprocessed feature vectors. As tasks grew more diverse, specialized corpora were introduced to address domain-specific needs. FIRE [129], an annual evaluation series, focused on Indian languages (e.g., Hindi, Bengali) and legal retrieval, providing critical resources for underrepresented languages despite being smaller than TREC or NTCIR. Similarly, domain-specific resources such as MEDLINE/MEDLARS and TREC Genomics [71] for biomedical search, the TREC Legal Track [37] for jurisprudence, and the TREC Microblog Track [167] for social media retrieval demonstrated the importance of tailoring corpora to task requirements, often using recall-oriented measures (e.g., in legal search).

7.1.4 ***Modern Heterogeneous & Synthetic Benchmarks.*** With the advent of neural retrieval, synthetic data generation emerged as a viable strategy. InPars [16] leveraged large language models offline to generate question–document pairs from a small set of annotated examples. These pairs were filtered by likelihood scores, and the top examples were used to fine-tune smaller, more efficient retrievers such as monoT5 within a multi-stage pipeline. This approach reduced reliance on costly LLM inference during retrieval, while scalable synthetic data enabled dense retrievers to be trained effectively. The authors also experimented with prompt designs to improve question quality and tested the method across multiple datasets.

Building on this, modern retrieval benchmarks emphasize heterogeneity, robustness, and generalization. BEIR [174] unified 18 public retrieval datasets spanning nine task types, including fact checking, citation prediction, duplicate question detection, argument retrieval, news retrieval, question answering, tweet retrieval, biomedical IR, and entity retrieval. It has become one of the most comprehensive platforms for testing zero-shot generalization of dense, sparse, re-ranker, and hybrid systems, while also demonstrating that traditional lexical models like BM25 remain strong baselines. In a similar spirit, MTEB [132] expanded evaluation to more than 50 datasets across tasks such as classification, bitext mining, and retrieval. While not strictly a RAG benchmark, MTEB includes retrieval tasks such as SciFact [182], FiQA [121], and HotpotQA [205], allowing systematic comparison of embedding models and retrievers for both general-purpose and specialized domains. Efforts toward unification and scalability have produced benchmarks such as KILT [143], which consolidates 11 datasets across fact checking, entity linking, slot filling, open-domain QA, and dialogue

into a single knowledge source of 5.9 million articles with provenance annotations. This allows joint evaluation of retrieval and generation while ensuring evidence traceability.

7.1.5 ***Specialized QA & Robustness Benchmarks.*** Other datasets emphasize more specialized evaluation dimensions such as entity-centric reasoning, counterfactual scenarios, and robustness under noisy conditions. PopQA [122] probes long-tail factual knowledge with around 14,000 Wikidata-based questions spanning 16 relation types (e.g., occupation, place of birth, capital of, author). By sampling subject–relation–object triples and converting them into natural language questions, it enables precise analysis of entity popularity and relation type while highlighting the difficulty of retrieving obscure facts. Unlike benchmarks dominated by well-known entities, PopQA stresses the long-tail distribution, making it valuable for testing both parametric memory in LLMs and the effectiveness of retrieval augmentation. Complementing this, IfQA [207] introduces over 3,800 counterfactual QA pairs framed with "if" clauses, requiring reasoning about hypothetical scenarios rather than factual recall. The dataset covers entities, dates, numbers, and reasoning-based responses, with supervised and few-shot settings that reveal the difficulty current QA systems face in counterfactual reasoning. Similarly, IFIR [169] targets instruction-following retrieval in expert domains like finance, law, and healthcare. With 2,426 examples validated by experts, it introduces multi-level instruction complexity and an evaluation metric, INSTFOL, for measuring adherence to instructions. Results across 15 retrievers show that instruction-tuned models are more robust but still struggle as instruction complexity increases, underscoring the need for domain-specific training data. Finally, robustness to adversarial or noisy retrieval has become a central focus. The MIRAGE dataset [141] provides 37,800 document chunks with positive and negative samples designed to simulate adversarial retrieval attacks. By testing whether RAG systems can remain faithful and consistent when exposed to misleading passages, MIRAGE highlights vulnerabilities in retrieval resilience and provides a benchmark for developing models capable of filtering noise while maintaining accurate answers.

Retriever-focused corpora provide the foundation for evaluating retrieval effectiveness, generalization, and resilience across diverse IR and RAG scenarios.

7.2 Generator-Focused Benchmarks

7.2.1 ***Summarization Benchmarks Benchmarks:*** Summarization has been one of the most prominent areas for evaluating text generation, and several large-scale benchmarks have been developed to capture different aspects of this task. One of the earliest and most widely used resources is the CNN/Daily Mail dataset [70], which contains over 287,000 news articles paired with human-written highlights. Each article is accompanied by short bullet-point summaries, making it a cornerstone benchmark for both extractive and abstractive summarization models. Building on this, the XSum (Extreme Summarization) dataset [133] introduced by BBC researchers pushes models toward more abstractive capabilities. With around 226,000 BBC articles paired with highly condensed, single-sentence summaries, XSum requires systems to move beyond simple copying and instead generate concise paraphrases that capture the core meaning. This makes it significantly more challenging than CNN/Daily Mail and is a popular benchmark for testing the abstractive abilities of summarization models. To provide greater diversity, the NewsRoom dataset [64] extends summarization evaluation to a much larger scale, with 1.3 million news articles paired with human-written summaries from 38 publishers spanning nearly two decades (1998–2017). Unlike earlier benchmarks, NewsRoom explicitly captures variation in summarization style—ranging from extractive to abstractive to mixed strategies. Summaries, often written for search engines and social media, reflect real-world practices, and the dataset has been analyzed through measures such as coverage, density, and compression, offering a more comprehensive training and evaluation resource.

Dataset	Type	One-Sentence Description
CNN/Daily Mail	News Summarization	287K news articles with human-written highlights, widely used for extractive and abstractive summarization.
XSum	Abstractive Summarization	226K BBC articles paired with single-sentence abstractive summaries requiring paraphrasing beyond copying.
NewsRoom [64]	Diverse Summarization	1.3M news articles with summaries from 38 publishers, covering extractive, abstractive, and mixed styles.
SummEval [45]	Summarization Evaluation	Human-annotated benchmark where system outputs are rated on coherence, consistency, fluency, and relevance.
KdConv [221]	Knowledge-Grounded Dialogue	4.5K Chinese dialogues (86K utterances) in film, music, and travel domains with knowledge-triple annotations.
DSTC7-ADV	Dialogue Robustness Evaluation	Dialogue evaluation sets with adversarial examples to stress-test dialogue system robustness.
Topical-Chat [63]	Knowledge-Grounded Dialogue	11K human-human conversations over 8 diverse topics with symmetric and asymmetric knowledge settings.
PersonaChat / RealPersonaChat (RPC) [202]	Persona-Grounded Dialogue	14K Japanese dialogues where speakers use their own personas with demographic and psychological traits annotated.
DailyDialog [106]	Everyday Dialogue	13,118 dialogues across 10 daily-life themes, annotated for dialogue acts and emotions.
DSTC10 (LLM-Eval) [156]	Dialogue Benchmark	Dialogue State Tracking Challenge dataset with tasks and evaluation sets for LLM-based dialogue evaluation.
WMT22 MQM Test Set [52]	Machine Translation Evaluation	Human-annotated test set using MQM for English–German, English–Russian, and Chinese–English translations.
Wizard of Wikipedia (WoW) [40]	Knowledge-Grounded Dialogue	22,311 dialogues (202K utterances) grounded in Wikipedia passages across 1,365 topics.
GSM8K [36]	Mathematical Reasoning	8.5K grade-school math word problems requiring 2–8 steps of reasoning in natural language.
e-SNLI [18, 22]	Explainable NLI	Extension of SNLI with natural language explanations for entailment, contradiction, and neutral labels.
DROP [42]	Discrete Reasoning QA	96K questions requiring arithmetic, counting, and comparisons over paragraphs, mostly from Wikipedia.
Cosmos QA [77]	Commonsense Reading Comprehension	35,588 multiple-choice questions from 21K personal narratives requiring contextual commonsense reasoning.
WikiEval [44]	Context/Answer Evaluation	50 edited Wikipedia pages with varied contexts and three answer types to study evaluation effects in RAG.
GLUE [184]	General NLU Benchmark	Nine sentence understanding tasks (classification, inference, similarity) with a diagnostic test set.

Table 6. RAG Generator-Focused Corpora

Dataset	Type	One-Sentence Description
SuperGLUE [183]	Advanced NLU Benchmark	Successor to GLUE with eight more challenging tasks emphasizing low-resource and transfer learning.
AIS Benchmark [126]	Trajectory Prediction	200 vessel trajectories (403K AIS records) with detailed ship speed, direction, and movement attributes.
OpinSummEval [165]	Opinion Summarization Evaluation	Benchmark for summarizing noisy, contradictory reviews with focus on aspects and sentiments.

Table 7. RAG Generator-Focused Corpora

Beyond large-scale datasets, benchmarks have also been developed with a focus on human evaluation. SummEval [45] provides a human-annotated benchmark for evaluating abstractive summarization systems, where multiple system outputs are rated for coherence, consistency, fluency, and relevance. By directly incorporating human judgments, SummEval helps address limitations of automatic metrics and serves as a gold standard for correlation studies. Finally, specialized datasets such as OpinSummEval [165] expand the scope of summarization evaluation to more complex domains. Designed for opinion summarization, it focuses on extracting aspects and sentiments from large collections of user reviews, which are often noisy and contradictory. This makes evaluation particularly challenging, as systems must distill useful insights from highly variable and sometimes conflicting opinions.

Together, these benchmarks reflect the evolution of summarization evaluation—from large-scale news corpora like CNN/Daily Mail and XSum, to diverse and style-rich resources such as NewsRoom, to human-centered evaluations like SummEval, and finally to specialized tasks like OpinSummEval. They collectively provide complementary perspectives for assessing the strengths and weaknesses of summarization systems.

7.2.2 ***Dialogue and Conversational Benchmarks:*** Dialogue and conversational benchmarks are crucial for evaluating systems that aim to replicate human-like interactions. These datasets vary widely in their focus, from knowledge-grounded conversations to daily life interactions, adversarial robustness, and persona-driven dialogue, providing a comprehensive testbed for advancing dialogue research.

One example is KdConv, a large-scale Chinese multi-domain dialogue dataset designed for knowledge-driven conversation [221]. It contains 4.5K dialogues (about 86K utterances) across domains such as film, music, and travel, with an average of 19 turns per dialogue—significantly longer than earlier datasets. Each utterance is annotated with related knowledge triples, enabling explicit modeling of knowledge planning, grounding, and adaptation. Benchmark experiments with both generation-based models (Seq2Seq, HRED) and retrieval-based models (BERT) show that incorporating knowledge improves performance, though models still struggle with producing knowledge-coherent, human-like conversations.

Building on the theme of robustness, the DSTC7-ADV dataset introduces adversarial dialogue examples—both open- and closed-ended—specifically to stress-test dialogue evaluators. Similarly, DSTC10 from the Dialogue State Tracking Challenge extends this line of work by providing dialogue tasks and evaluation sets that are widely used for benchmarking large language model (LLM)-based dialogue evaluation [156].

In contrast, Topical-Chat [63] focuses on open-domain, knowledge-grounded conversations. It consists of 11,000 human-human conversations spanning eight diverse topics such as movies, politics, sports, and science. Unlike datasets that restrict roles or narrow domains, Topical-Chat supports both symmetric and asymmetric knowledge settings, making the conversations more realistic and challenging.

Personalization is another key direction in dialogue benchmarks. PersonaChat and its extension, RealPersonaChat (RPC), advance this by modeling personality traits within conversation. RPC, a Japanese dialogue corpus with 14,000 conversations, goes beyond earlier datasets by using real personas rather than artificially assigned ones. It includes demographic and psychological attributes such as Big Five traits, capturing natural expressions of personality and yielding more authentic conversations, although with lower explicit persona frequency compared to JPersonaChat [202]. Analyses confirm its value for studying personality-driven dialogue and training more human-like systems.

To better reflect everyday human interactions, DailyDialog [106] offers 13,118 multi-turn dialogues, averaging around eight turns, each focused on a single topic across ten daily-life themes such as relationships, work, and ordinary life. Each utterance is annotated with dialogue acts and emotions, making it a valuable resource for modeling natural communication patterns. Studies show that incorporating these annotations improves dialogue quality in both retrieval- and generation-based models.

Finally, knowledge-grounded conversation is taken further with Wizard of Wikipedia (WoW) [40], which contains 22,311 dialogues and about 202,000 utterances across 1,365 topics including food, movies, history, and science. In these dialogues, a "wizard" uses retrieved Wikipedia passages to provide informed responses, while an "apprentice" plays the role of a curious learner. This explicit grounding in external knowledge makes WoW a key benchmark for testing retrieval-augmented dialogue systems.

Together, these benchmarks highlight the breadth of conversational evaluation—from adversarial robustness and persona-driven interaction to knowledge-grounded dialogue and daily-life communication—each pushing dialogue systems closer to natural, contextually aware, and human-like conversations.

7.2.3 ***Question Answering & Reasoning Benchmarks:*** Beyond retrieval, robust evaluation of RAG systems also depends on benchmarks that test a generator's ability to produce accurate, well-grounded, and logically consistent answers. Question answering (QA) and reasoning datasets are particularly valuable, as they challenge models not only to extract information but also to reason over multiple steps, infer unstated facts, and maintain factual grounding in diverse contexts.

GSM8K is a collection of 8.5K high-quality, linguistically diverse grade school math word problems [36]. Each problem typically requires 2–8 reasoning steps using basic arithmetic, with solutions expressed in natural language rather than equations. Designed around principles of high quality, diversity, and moderate difficulty, GSM8K has become a key benchmark for diagnosing reasoning weaknesses in language models and for evaluating methods such as finetuning and verification.

Building on this, the DROP dataset (Discrete Reasoning Over Paragraphs) pushes models toward more complex reasoning over text [42]. Comprising over 96,000 adversarially crowdsourced questions paired with passages, primarily from Wikipedia sports and history summaries, DROP requires operations like addition, subtraction, counting, sorting, and comparison across multiple mentions within a passage. Unlike span-based QA tasks, DROP demands discrete reasoning, making it a challenging benchmark for both comprehension and reasoning abilities.

Complementing numerical and logical reasoning, Cosmos QA evaluates contextual commonsense reasoning [77]. Built from over 21,000 personal narrative contexts collected from weblogs, it includes 35,588 multiple-choice questions that require models to "read between the lines." Instead of relying solely on explicit facts, models must infer causes, effects, motivations, situational details, and counterfactuals not directly stated in the passage. Validated through Amazon Mechanical Turk, the dataset reveals a significant gap between human performance (94% accuracy) and current state-of-the-art models, underscoring its difficulty and importance for advancing commonsense comprehension.

Finally, WikiEval [44] offers a compact but highly relevant resource for RAG evaluation. Built from 50 recently edited Wikipedia pages, it provides a question, a succinct ideal context, a longer redundant context, and three answer variants—grounded, ungrounded, and poor—for each page. This design enables systematic analysis of how models handle context selection, redundancy, and grounding, making WikiEval especially useful for assessing context relevance and answer quality in retrieval-augmented generation.

7.2.4 ***Natural Language Inference (NLI) and Explanation Benchmarks:*** Evaluating RAG generators often requires testing not only factual accuracy but also logical consistency and interpretability. Natural Language Inference (NLI) and explanation benchmarks provide valuable resources in this direction by assessing whether models can correctly capture entailment relationships and justify their reasoning in natural language.

e-SNLI is an extension of the widely used Stanford Natural Language Inference (SNLI) corpus [18], designed to enhance interpretability in inference tasks. While SNLI contains premise–hypothesis pairs labeled as entailment, contradiction, or neutral, e-SNLI [22] goes further by adding human-annotated natural language explanations for each relation. Explanations were crowd-sourced via Amazon Mechanical Turk, with annotators first highlighting important words and then writing full-sentence justifications. The dataset includes one explanation per training pair and three per validation and test pair, making it the largest collection of natural language explanations for NLI. By requiring models not only to predict the correct label but also to provide a supporting rationale, e-SNLI serves as a critical benchmark for testing explanation quality, an important dimension for RAG outputs where users demand transparency in reasoning.

7.2.5 ***General Language Understanding Benchmarks:*** Beyond inference, general-purpose language understanding benchmarks have become central to evaluating how well models handle a broad range of linguistic phenomena. These benchmarks are equally useful in RAG generator evaluation since they test robustness across diverse tasks such as classification, paraphrase detection, and reasoning—capabilities that directly affect answer quality in knowledge-intensive applications.

GLUE [184] is one of the most influential benchmarks for natural language understanding (NLU), combining nine English tasks across different domains and levels of difficulty. It covers single-sentence classification (e.g., sentiment analysis, linguistic acceptability), similarity and paraphrase detection (e.g., Quora Question Pairs, Semantic Textual Similarity), and inference tasks (e.g., MultiNLI, RTE, Winograd NLI). GLUE also includes a diagnostic dataset with carefully crafted examples targeting phenomena such as logic, world knowledge, and coreference. With its online leaderboard and standardized evaluation, GLUE has become a cornerstone for developing robust, general-purpose NLU systems.

Building on its success, SuperGLUE [183] was introduced as a more challenging successor, aimed at pushing the boundaries of NLU further. It consists of eight difficult tasks, including BoolQ, CB, COPA, MultiRC, ReCoRD, RTE, WiC, and WSC, chosen for their complexity relative to existing models while remaining solvable by humans. Unlike GLUE, SuperGLUE emphasizes low-resource conditions, with most tasks offering fewer than 10,000 training examples, thereby encouraging research into transfer learning, multitask approaches, and self-supervised methods. It also provides human performance baselines, standardized splits, and a public leaderboard, making it one of the most rigorous testbeds for advancing model generalization.

Together, e-SNLI, GLUE, and SuperGLUE enrich RAG generator evaluation by assessing logical consistency, explanation quality, and broad language understanding. These benchmarks ensure that evaluation goes beyond surface-level correctness, probing deeper into reasoning, interpretability, and robustness across diverse NLP tasks.

7.2.6 ***Domain-Specific Benchmarks:*** In addition to general-purpose evaluation datasets, domain-specific benchmarks play a crucial role in testing RAG systems and language models under specialized conditions. These resources capture unique challenges—such as technical vocabulary, structured signals, or safety-critical reasoning—that are often absent in broader NLP benchmarks.

AIS Benchmarks [126] represent one such domain-specific dataset, designed for maritime trajectory prediction and data mining. The benchmark contains 200 clean vessel trajectories with over 403,000 AIS (Automatic Identification System) records, each annotated with attributes such as latitude, longitude, speed over ground (SOG), course over ground (COG), rate of turn (ROT), time, and vessel ID (MMSI). Dataset construction involved preprocessing raw AIS data, selecting valid routes of sufficient length and complexity, cleaning noisy trajectories, and interpolating missing values. Covering diverse vessel types—including cargo ships, tankers, and military vessels—the dataset provides detailed distributions of vessel speed and direction. It has become a standard benchmark for evaluating maritime anomaly detection, path prediction, and other data-mining algorithms, demonstrating the importance of domain-specific corpora for advancing specialized applications.

7.2.7 ***Machine Translation & Text Quality Benchmarks:*** Another key strand of evaluation focuses on translation and linguistic quality, where human alignment and multilingual coverage are critical. The WMT22 MQM test set [52], introduced at the 2022 Workshop on Machine Translation, exemplifies this by providing a benchmark for machine translation annotated using Multidimensional Quality Metrics (MQM). The dataset includes high-quality human evaluations across several language pairs, specifically English–German, English–Russian, and Chinese–English. Unlike automatic overlap-based metrics, MQM offers a more fine-grained framework by capturing multiple error types such as adequacy, fluency, and accuracy. This makes it a valuable benchmark not only for advancing machine translation but also for informing RAG evaluation on dimensions of text quality and linguistic precision.

7.3 End-to-End System Dataset

To holistically evaluate RAG systems and other context-grounded generation models, several benchmarks have been introduced that test not only retrieval or generation in isolation but also the full end-to-end pipeline of evidence retrieval, integration, and response generation. These datasets are designed to capture the complexity of real-world scenarios, where factual correctness, faithfulness, and robustness to noise must be jointly assessed.

FEVER [176] is one of the earliest and most influential large-scale benchmarks for claim verification against textual sources. It contains 185,445 human-generated claims derived and mutated from Wikipedia sentences, each labeled as SUPPORTED, REFUTED, or NOT ENOUGH INFO. For claims in the first two categories, annotators also identified exact sentences serving as supporting or refuting evidence. As a result, FEVER has become a cornerstone dataset for advancing fact-checking, evidence retrieval, and automated verification methods.

Building on this tradition, MIRAGE [141] was specifically designed for evaluating Retrieval-Augmented Generation (RAG) systems under noisy and adversarial conditions. It consists of 7,560 question–answer pairs mapped to a retrieval pool where each query is paired with at least one oracle (relevant) chunk and several distractor (irrelevant) chunks. By balancing relevance, noise, and domain coverage, MIRAGE provides a compact yet challenging benchmark that stresses the resilience of RAG systems to misleading context.

Complementing these task-oriented datasets, ContextualJudgeBench [198] focuses on evaluating the evaluators themselves—that is, large language model (LLM) judges operating in contextual settings such as RAG. Unlike prior judge benchmarks that primarily assess responses without context, ContextualJudgeBench emphasizes evaluation

grounded in external evidence, where criteria like refusal, faithfulness, and completeness interact. This makes it a valuable resource for advancing the reliability and interpretability of LLM judges in real-world contextual evaluation.

Building upon these foundational benchmarks, the research landscape has seen a rapid expansion of specialized datasets designed to address more specific RAG challenges. For domain-specific evaluation, for instance, OmniEval [186] was introduced for the financial domain and CDQA [201] for dynamic Chinese question answering. To address more complex reasoning patterns, MultiHop-RAG [172] provides a benchmark for multi-hop queries, the CRUD benchmark [119] for categorizing different RAG tasks, and MTRAG [89] for multi-turn conversational RAG. Other benchmarks aim for broader, more realistic testing, such as the large-scale, industry-focused RAGBench [53], and the Comprehensive RAG Benchmark (CRAG) [203], which is designed to reflect the diversity and dynamism of real-world questions.

Together, these foundational benchmarks—FEVER for fact verification, MIRAGE for RAG robustness, and ContextualJudgeBench for judge evaluation—along with the numerous specialized datasets for multi-hop, multi-turn, and domain-specific challenges, offer a rich and expanding toolkit for comprehensive end-to-end system assessment. They highlight not only the difficulty of combining retrieval and generation but also the importance of reliable evaluation frameworks that keep pace with increasingly complex RAG pipelines.

8 Challenges of Existing RAG Evaluation

Despite the rapid advancements in Retrieval-Augmented Generation, the methodologies for evaluating these systems face significant and multifaceted challenges [3, 8, 10]. These limitations span the entire RAG pipeline, from fundamental issues in information retrieval to the complexities of assessing generated text and the holistic behavior of the end-to-end system. This section will systematically examine these challenges to highlight critical gaps and guide future research toward more robust and reliable evaluation. Critically examining these limitations is essential for practitioners to avoid being misled by imperfect metrics and for researchers to develop the next generation of evaluation techniques.

8.1 Retriever-Centric Challenges

Many of the fundamental challenges in retriever evaluation stem from the limitations of the training data and benchmarks used. A major issue is the training–inference mismatch and sampling bias, where dense retrievers are trained on limited or easy negative samples (e.g., in-batch negatives), which fails to reflect the complexity of retrieving from a full corpus with diverse distractors. Compounding this is the incompleteness of relevance labels in most benchmark datasets; systems are often unfairly penalized for retrieving valid but unannotated documents, which underestimates true performance. Furthermore, most retrieval research is benchmarked on English-centric corpora creating significant domain and language restrictions that limit generalizability to specialized or multilingual settings. These issues are exacerbated by biases in dataset construction, as many benchmarks were originally designed for lexical retrieval, inadvertently favoring older term-matching systems over modern dense retrievers that rely on semantic similarity.

Beyond the data, significant challenges are rooted in the intrinsic architectural limitations of the retriever models themselves. Some approaches rely on static document representations, such as freezing embeddings during training to reduce computational cost, which in turn prevents fine-grained adaptation to new domains or tasks. A more fundamental issue is the lack of phrase-level modeling, as many retrievers treat documents as bags of words or single-term embeddings, overlooking the compositional meaning carried by multi-word expressions. This leads to the dual problems of vocabulary mismatch, where lexical models fail to capture paraphrases, and semantic drift, where dense models risk over-generalization. Finally, a persistent challenge is the poor interpretability of dense retrieval; the opaque

nature of dense similarity scores makes it difficult to explain why a particular passage was retrieved, which severely hampers error analysis and debugging.

Several other challenges arise at the system and pipeline level, concerning the retriever's practical deployment and integration. Current evaluation metrics often suffer from context insensitivity, assuming a static query-document match while ignoring dynamic factors like user intent or evolving information needs. Scalability is a pressing practical concern, as dense retrieval models demand substantial computational resources and high-memory hardware, raising the barrier for deployment. This also introduces scalability trade-offs, as billion-scale indices require Approximate Nearest Neighbor (ANN) search or sharding, which forces a compromise between recall and latency. Architecturally, the community's reliance on the fixed retrieval–reranking pipeline can limit efficiency and prevent deeper, end-to-end optimization of the retrieval and generation processes. This contributes to a lack of robustness to out-of-domain queries, where performance often collapses in specialized settings, and perpetuates bias and fairness concerns, as training on unrepresentative corpora can amplify viewpoint imbalances in the final generated outputs.

8.2 Generator-Centric Challenges

While retriever challenges focus on finding the right information, Generator-Centric Challenges center on the complex task of assessing the quality, faithfulness, and safety of the generated text. These challenges are multifaceted, ranging from the inherent limitations and biases of both fine-tuned and general-purpose evaluators to the system's vulnerability to adversarial attacks. Understanding these shortcomings is critical for developing reliable evaluation frameworks, especially in high-stakes settings where factual accuracy and trustworthiness are paramount.

8.2.1 ***Limitations of Fine-Tuned Judge Models:*** Fine-tuned judge models often achieve strong results on their in-domain test sets but show important limitations compared to proprietary models like GPT-4, restricting their general applicability for Large Language Model evaluation [76]. These shortcomings stem largely from overfitting during the fine-tuning process, which transforms a general-purpose foundation model into a rigid, task-specific classifier. As a result of this specialization, fine-tuned judges struggle to generalize when applied to evaluation schemes different from their training setup, such as attempting pairwise selection after being trained on pointwise grading, or shifting from single-turn to multi-turn evaluation.

This rigidity leads to several specific performance failures. Fine-tuned judges are prone to superficial biases, frequently favoring answers that appear more formal or verbose even when factually incorrect—an issue not observed in GPT-4. Moreover, they lack the flexibility for aspect-specific evaluation on dimensions such as factuality, toxicity, or safety. They also do not benefit from prompt-engineering strategies like in-context learning or chain-of-thought, sometimes even performing worse under these conditions. Ultimately, their generative capabilities do not translate into broader evaluation competence, leaving them closer to rigid classifiers than adaptive evaluators. For these reasons, fine-tuned judge models cannot currently substitute for proprietary systems like GPT-4 in general evaluation scenarios, and their use should be approached with caution in real-world applications.

8.2.2 ***Bias and Preference Leakage:*** One of the most pressing challenges in generator-centric evaluation is the prevalence of bias and preference leakage in "LLM-as-a-Judge" systems. Preference leakage [101] occurs when the same or related Large Language Models are used both to generate synthetic training data and to act as judges, causing the evaluation to become systematically biased. These judges tend to favor student models trained on their own outputs, unfairly inflating performance in a self-reinforcing loop that undermines fair comparison. This is closely related to

self-enhancement or self-favoring bias, where evaluators generally prefer outputs generated by their own family of models, in some cases even rating Large Language Model-generated text more highly than human-written text.

Beyond biases related to the evaluation setup, Large Language Model judges also exhibit a wide range of inherent behavioral and cognitive biases. The most commonly cited are positional bias, where the order of candidates influences the score (e.g., GPT-4 tends to favor the first response, while ChatGPT favors the second [185]), and verbosity bias, the tendency to reward longer, more elaborate answers even if they are factually incorrect. Broader benchmarking has revealed substantial susceptibility to numerous other cognitive biases, including compassion fade, egocentric bias, salience bias, bandwagon effects, and attentional bias, with studies finding that nearly 40% of evaluations are affected and that alignment with human rank-order judgments can be as low as 44% [92]. Furthermore, judges have shown vulnerability to misinformation oversight, gender bias, authority bias, beauty bias, and demographic biases, many of which have been found to be exploitable through prompt-based attacks [26, 220].

These systems also display more fine-grained vulnerabilities and are compounded by the inherent subjectivity of evaluation. Experiments show that judges can systematically penalize epistemic markers (e.g., "I'm not sure"), unfairly lowering scores for otherwise correct outputs, a behavior not seen in human judges [97]. Comparative benchmarks revealed that even the strongest systems like GPT-4 and Llama-3 70B fall short of inter-human agreement and struggle with prompt sensitivity, leniency bias, and evaluating partially correct answers [173]. This is exacerbated by their limited reasoning and mathematical capabilities. Ultimately, the subjective nature of judging qualities like coherence and relevance varies widely across annotators and cultures, making consistency difficult. Taken together, this body of work underscores that while Large Language Models offer scalable evaluation, their judgments remain vulnerable to systematic bias, leakage, and manipulation. Addressing these shortcomings will require robust calibration methods (such as balanced position calibration and human-in-the-loop verification), stronger benchmark design, and a balanced integration of human oversight.

8.2.3 ***Vulnerability to Adversarial Attacks:*** A growing concern in generator-centric evaluation is the vulnerability of "LLM-as-a-Judge" systems to adversarial attacks and manipulations. As these systems are increasingly deployed for automated benchmarking and safety evaluation, their robustness must be checked against subtle attacks that can distort outcomes. One major line of research highlights backdoor-style adversarial attacks, where evaluators are deliberately manipulated through data poisoning to unfairly boost a model's scores or penalize competitors [177]. Studies show that even minimal data poisoning—as little as one percent—can drastically inflate evaluation results, with attacks categorized by escalating levels of access: web poisoning, malicious annotators during fine-tuning, and direct weight poisoning. While defenses are difficult, lightweight strategies such as model merging have proven effective, reducing attack success rates to nearly zero while maintaining state-of-the-art performance.

Beyond data poisoning, evaluators are also vulnerable to manipulations at inference time. Research demonstrates that judges can be deceived by simple universal adversarial triggers—short, crafted phrases appended to a text that are sufficient to trick an evaluator into assigning an inflated score, regardless of the underlying quality [149]. These triggers can be learned on smaller models and successfully transferred to larger ones, with absolute scoring methods being far more vulnerable than comparative assessments. In a more subtle form of manipulation, evaluation systems are also susceptible to superficial artifacts that distort judgment. An investigation into automated safety judges revealed that features like apologies, added authority, verbosity, or "halo" phrasing could skew results by up to 98 percent [27]. This study also found that larger models were not consistently more robust and that judgments could be inconsistent even with deterministic settings, raising concerns about reproducibility. As a partial mitigation, jury-based evaluations that

aggregate outputs from multiple models have been shown to improve robustness, but they still fail to fully eliminate artifact sensitivity.

Taken together, these findings demonstrate that adversarial attacks and stylistic artifacts pose serious risks to the reliability of Large Language Model-based evaluators. Even small manipulations—whether through poisoned data, crafted prompts, or superficial phrasing—can distort judgments, undermining the fairness, safety, and reproducibility of the entire evaluation process. Addressing these vulnerabilities requires the development of more robust defenses, the use of diversified evaluation methods like comparative assessment, and the creation of artifact-aware benchmark designs to ensure the trustworthy deployment of "LLM-as-a-Judge" systems in real-world settings.

8.2.4 ***Shortcomings of Human and LLM Evaluators:*** Despite their increasing adoption, "LLM-as-a-Judge" systems exhibit substantial shortcomings that limit their reliability, as they are not yet at the level of human evaluators in terms of consistency and nuanced judgment. A primary challenge lies in fact-sensitive tasks, where reference-free evaluators often fail to verify factual correctness in knowledge-intensive domains because they lack the necessary external knowledge, instead rewarding fluent but inaccurate text. Another limitation is insufficient score differentiation, as judges tend to collapse their outputs into low-variance ratings (e.g., consistently assigning "3" on a 1–5 scale), making it difficult to distinguish between high-performing systems. These evaluators are also highly sensitive to prompting, where slight variations in phrasing or candidate ordering can lead to large swings in outcomes, undermining reproducibility. Furthermore, they suffer from high computational cost, with proprietary models incurring significant API expenses, and limited interpretability, as single numerical scores provide little actionable feedback for system improvement.

Importantly, these shortcomings are not restricted to automated systems; human evaluators themselves exhibit serious weaknesses when tasked with assessing modern language models. Research shows that non-expert evaluators perform at near-random chance when distinguishing outputs from models like GPT-3.5 from human-authored text [33]. This is because they often rely on outdated, surface-level cues such as grammar or style, which are no longer reliable indicators of machine authorship, while frequently overlooking deeper content-based errors like factual inaccuracies, contradictions, or subtle reasoning failures. Studies have found that human evaluators often give contradictory justifications for their ratings and hold misconceptions about machine capabilities, leading to low inter-annotator agreement. These findings suggest that traditional small-batch human evaluations, often run on crowd-sourcing platforms, are insufficient for accurately assessing today's highly fluent Large Language Models.

This persistent gap between automated and human evaluation is evident even when using advanced metrics. For instance, on human-annotated opinion summarization benchmarks, neural-based alternatives like BERTScore and BARTScore show stronger alignment with human ratings than traditional metrics like ROUGE, but they remain inconsistent across different evaluation dimensions [165]. In summary, both LLM and human evaluators fall short in critical ways: Large Language Model judges suffer from factual unreliability, bias, limited granularity, and high cost, while human evaluators often misjudge machine-generated text due to reliance on outdated heuristics and lack of consistency. Together, these shortcomings highlight the urgent need for hybrid approaches that combine human expertise with robust, domain-aware automatic metrics, alongside better training, calibration, and interpretability mechanisms for evaluation frameworks.

8.2.5 ***Reliability and Robustness in High-Stakes Settings:*** As Large Language Models are increasingly deployed as evaluators in safety-critical domains such as content moderation, red-teaming, and guardrailing, their own reliability and robustness have become a pressing concern. While these systems promise scalable evaluation, research reveals

that they are highly vulnerable to distribution shifts, prompt sensitivity, and adversarial manipulation. For instance, a key study investigating widely used safety benchmarks like HarmBench [127] and LLaMA Guard [80] highlighted two critical challenges: inconsistent judgments "in the wild" due to minor stylistic variations in text, and adversarial attacks that explicitly target the judge rather than the generator [43]. Experiments showed that small stylistic changes could increase false negative rates by up to 0.24 (causing harmful content to be misclassified as safe), while adversarially crafted responses could lead safety judges to misclassify nearly all unsafe content. This reveals that high baseline performance scores often create a false sense of robustness.

These findings highlight that Large Language Model judges, while promising, are not yet robust enough for deployment in critical settings without significant safeguards, as they often inherit the systematic weaknesses of their underlying models. Addressing these weaknesses will require a multi-pronged approach. First, there is a need for more rigorous, diversified benchmarking that explicitly tests for robustness under distribution shifts and adversarial perturbations. Second, methodological advances are required, such as jury-based evaluation (aggregating judgments from multiple models), calibration techniques to mitigate bias, and hybrid human-in-the-loop oversight. Without these improvements, Large Language Model-based evaluators risk amplifying errors and offering unreliable assurances of safety in domains where precision, accountability, and a holistic threat model are essential.

8.2.6 ***Broader Evaluation Gaps:*** Several broader evaluation gaps persist that extend beyond specific component limitations, stemming from fundamental issues with data, metrics, and subjectivity. When entirely new services are built with Large Language Models, there is often no existing data for reliable evaluation. While synthetic data generation is a proposed solution, it must be applied cautiously to complement human annotators rather than replace them, to avoid reinforcing model artifacts. Even with data, disagreements among experts highlight the subjective nature of what constitutes a "good" summary or response. This is reflected in our metrics, which struggle to capture the fine-grained distinctions between "good" and "excellent" outputs. Traditional metrics like BLEU or ROUGE are fundamentally limited to token overlap and correlate poorly with human judgments on semantic qualities like coherence and factual grounding. While embedding-based approaches like BERTScore or COMET improve on semantic similarity, they often lack interpretability, failing to explain why one output is better than another.

Beyond the metrics, challenges arise from the inherent weaknesses of the Large Language Models themselves and the pipelines used to evaluate them. Models frequently hallucinate, struggle with reasoning consistency, and suffer from knowledge recency issues; without rigorous evaluation, these errors can go undetected and lead to harmful real-world consequences. Empirical work has further exposed gaps in evaluation pipelines, identifying constraints such as limited prompt tuning and the absence of techniques like in-context learning or chain-of-thought prompting that could improve robustness. This research also highlights the critical bias introduced when poorly-aligned automatic metrics are used to pre-filter outputs for human evaluation, as well as the limited generalizability of studies constrained by time and budget.

Together, these challenges underscore the existence of broad evaluation blind spots where our current methods fall short. Metrics still fail to adequately capture hallucinations, subtle factual errors, or nuanced quality differences, while human annotators, though more reliable in principle, remain inconsistent and resource-intensive. The path forward requires combining traditional metrics, embedding-based methods, and "LLM-as-a-Judge" approaches with more rigorous human oversight. It also demands the development of new evaluation frameworks capable of better addressing subjectivity, grounding, and explainability. Without closing these gaps, our evaluations risk being shallow proxies rather than accurate reflections of true system performance.

8.3 Challenges on Total (End to End) System

Cross-Cutting Challenges: Despite progress in retrieval-augmented generation, evaluating end-to-end systems remains fraught with cross-cutting challenges that span both retriever and generator components. Unlike isolated retriever or generator evaluation, holistic assessment requires accounting for training–inference gaps, benchmark design trade-offs, resource limitations, and broader systemic biases. These issues collectively highlight why RAG evaluation is still fragmented and why progress demands frameworks that treat retrieval and generation as interdependent processes.

Training–Inference Misalignment: A central difficulty in evaluating dense retrievers is the Training–Inference Misalignment, which arises from the discrepancy between the conditions under which a model is trained and deployed. During training, retrievers often learn to distinguish relevant passages from a limited set of relatively "easy" negative samples, such as in-batch negatives or those selected by BM25. At inference, however, the retriever must search a full corpus containing many “hard negatives”—passages that are contextually similar but ultimately misleading or incomplete. This mismatch frequently leads to inflated performance scores during offline evaluation but results in poor real-world generalization. For example, a retriever may appear strong on a benchmark yet consistently fail to retrieve factually grounded evidence in a production environment. Addressing this challenge requires training strategies that more closely approximate the complexity of inference conditions, such as incorporating harder negatives, using adversarial distractors, or applying curriculum-based sampling.

Dynamic vs. Static Benchmarks: Another challenge lies in the choice between static and dynamic evaluation settings. Static benchmarks, such as MS MARCO or Natural Questions, are reproducible and allow for fair system comparison, but they quickly become outdated—especially for domains requiring knowledge freshness (e.g., news or biomedical updates). Dynamic benchmarks, such as live web retrieval or streaming corpora, preserve timeliness but sacrifice reproducibility, since the source corpus evolves over time. This trade-off between freshness and repeatability complicates fair benchmarking of RAG systems, particularly for applications where knowledge evolves rapidly. Research efforts are beginning to explore hybrid strategies, such as “living benchmarks” with periodic refreshes, but standardization remains elusive.

Unified Metrics: Current evaluation frameworks still lack a single, holistic metric that captures both retrieval effectiveness and generation quality. Retrievers are commonly assessed using Recall@k or nDCG, while generators are judged with metrics like ROUGE, BLEURT, LLM as a Judge or human ratings. This separation obscures how well retrieved evidence actually supports the generated answer. Without unified metrics, system optimization often prioritizes one component at the expense of the other, making it difficult to compare architectures in a fair, end-to-end manner. Recent attempts, such as combined faithfulness and coverage measures in RAGAS [44], hint at solutions, but these rely on a single evaluator model, which can introduce systematic bias. Developing reliable composite metrics that balance retrieval coverage, grounding quality, and answer correctness remains an open frontier.

Resource Constraints: Beyond theoretical performance, the practical deployment of Retrieval-Augmented Generation systems faces severe Resource Constraints related to hardware and operational bottlenecks. These challenges manifest in three key areas. First, memory requirements are often prohibitive, as the combined demand of storing large embedding indexes, handling long-context windows, and running inference can exceed Graphics Processing Unit (GPU) or Tensor Processing Unit (TPU) limits, especially in consumer or multilingual applications. Second, Latency is equally critical, as the cumulative time taken for both the retrieval and generation stages can make interactive systems feel sluggish. Finally, significant privacy concerns emerge when external indexes contain sensitive or proprietary documents, raising complex compliance and security challenges. Therefore, successfully optimizing across the competing demands

of memory footprint, inference speed, and privacy guarantees is a key determinant of whether Retrieval-Augmented Generation systems can scale to production workloads.

Evaluator and Framework Limitations: Even advanced automated evaluation frameworks have inherent shortcomings that can limit their practical application, particularly concerning data, hardware, and language support. Many systems like ARES [158] still require a calibration step using human-annotated data, which can be prohibitively costly in specialized domains such as law or medicine. Furthermore, the significant hardware demands for generating synthetic data and fine-tuning lightweight judge models can restrict accessibility for smaller organizations. A major constraint is also language coverage, as most prominent evaluation frameworks remain limited to English, with little validation performed in multilingual contexts.

Further limitations stem from the narrow scope and simplified assumptions common in current benchmark and dataset design. For example, many datasets only test simplified Retrieval-Augmented Generation architectures (such as prepending retrieved passages to the prompt), leaving the effectiveness of more complex strategies like Active Retrieval-Augmented Generation or retrieval-augmented planning untested. These scope restrictions weaken the generalizability of current benchmarks. A framework that focuses narrowly on multiple-choice Question Answering, for instance, makes it unclear whether its results are relevant to more complex, high-stakes tasks such as claim verification, rationale generation, or multi-turn dialogue support.

Community and Cost Considerations: Finally, the ecosystem still lacks shared infrastructure for evaluating full RAG pipelines. Leaderboards exist for isolated retriever or generator tasks, but few track end-to-end RAG performance in a consistent manner. Reliance on a single evaluation model, such as GPT-3.5-turbo in RAGAS, introduces systematic bias and limits generalizability. Moreover, high computational costs—both in GPU inference and API usage—make large-scale benchmarking impractical for many research groups. Distilled judges, such as smaller models approximating GPT-4, offer a cost-efficient alternative, but their reliability is still under question. Ensuring community-wide reproducibility will likely require open leaderboards, shared calibration datasets, and evaluation protocols that explicitly balance cost with accuracy.

Taken together, these challenges underscore that RAG evaluation cannot be fully solved by piecemeal advances in retriever metrics, generator judges, or benchmark design. Instead, holistic approaches that jointly optimize retrieval, generation, and evaluation—while accounting for cost, language diversity, and evolving knowledge—are essential for reliable end-to-end assessment.

9 Best-Practice Guidelines and Future Work

Having systematically examined the challenges and limitations of current evaluation methodologies, this section outlines the best practices and future research directions needed to address these gaps. The following guidelines and avenues for future work aim to pave the way for more robust, reliable, and fair evaluation frameworks for Retrieval-Augmented Generation. Pursuing these goals is essential for moving beyond the current state-of-the-art and enabling the development of more trustworthy and effective systems [9].

9.1 Retriever Centric

Future work in retriever-centric evaluation is focused on enhancing core model capabilities and broadening their applicability. One important direction is the joint optimization of query and document encoders for tighter alignment between representation spaces. Research must also advance semantic composition with techniques like phrase-level embeddings to capture nuanced relevance beyond simple token matches. To better assess modern architectures, unified

evaluation frameworks for hybrid sparse-dense systems are needed to fairly balance lexical precision with semantic recall. Furthermore, evaluation must extend beyond traditional ad-hoc search to more diverse and complex tasks such as open-domain Question Answering, conversational search, and multi-hop reasoning, while also expanding into specialized medical, legal, and multilingual domains.

Another key priority is to make evaluation more dynamic, practical, and user-centric. This includes developing context-aware and personalized evaluation that incorporates user intent and interaction feedback (such as clicks and dwell time) to better reflect real-world utility. To overcome the limitations of static test collections, future work should embrace denser and dynamic judgments through active learning or continual updating, and create temporal and evolving benchmarks that measure long-term reliability against knowledge decay. For systems designed for continual learning, new metrics are required to capture model staleness, update cost, and catastrophic forgetting. Finally, progress in explainable neural Information Retrieval, which provides justifications for retrieval decisions, and a push for cross-modal and multilingual evaluation are essential for improving trust and extending retrieval beyond text-only, English-centric paradigms.

To mitigate current challenges, several best practices should be adopted for a more holistic and robust assessment. Practitioners should use composite scorecards that integrate quality metrics (like Normalized Discounted Cumulative Gain), efficiency measures (latency, memory usage), and training costs to balance accuracy with scalability. It is also critical to stress-test retrieval models for robustness, including cross-domain generalization and adversarial attacks. Instead of relying on a single score, a portfolio of metrics should be used, including a combination of Large Language Model-based judges and traditional Information Retrieval metrics to balance speed and interpretability. Finally, practical stability checks (such as re-running prompts multiple times) and continuous human calibration are vital, alongside a focus on bias mitigation through normalization, penalties, and systematic audits to ensure fairness.

9.2 For Generator

Future work in generator-centric evaluation emphasizes overcoming the known limitations of both human and "LLM-as-a-Judge" evaluators by designing more robust, bias-resistant, and context-aware methods. A foundational direction is to improve human evaluation practices. This includes providing evaluators with calibration examples, aligning their implicit criteria with task usefulness rather than just "humanlikeness", and embedding evaluation in interactive settings like collaborative writing. To reduce inconsistency, transparent reporting of evaluator training and standardized design practices are essential, as is correcting evaluator misconceptions about the capabilities of modern, highly fluent Large Language Models.

A second major focus is addressing the systematic weaknesses in current evaluation frameworks. Even advanced methods have persistent blind spots: token-overlap metrics like BLEU and ROUGE ignore factual grounding, while embedding-based metrics such as BERTScore or COMET capture semantic similarity but lack interpretability. The inherent subjectivity of judging qualities like coherence and readability remains a challenge, and LLM-based evaluators introduce their own issues, including order bias, verbosity preference, calibration drift, and inconsistent scoring variance. Furthermore, the high computational cost of high-quality evaluators like G-Eval [115] and the workflow complexity of alternatives like PAIRS limit their practical application.

To overcome these issues, future work must target robustness, scalability, and reliability. This includes stress-testing generators with adversarial inputs and developing more cost-effective LLM judges by distilling larger models into smaller, efficient evaluators. The creation of standardized benchmarks with broad domain and multilingual coverage is also critical for reproducibility. Methodologically, promising directions include ensemble evaluation, where multiple

LLMs are aggregated to mitigate single-model bias, and confidence calibration, which adds interpretable uncertainty bands to automated scores. The field is also moving toward enhancing evaluator explainability, where models not only score outputs but also explain weaknesses at the sentence level, and developing self-correcting Large Language Models that are fine-tuned to critique and revise their own outputs.

A fourth critical area is bias mitigation and the expansion of evaluation scope. This involves conducting detailed bias audits and applying mitigation techniques such as adversarial training, normalization, randomized output ordering, and pairwise ranking approaches like Direct Preference Optimization [147]. Specific fixes, such as penalties for verbosity bias or using third-party evaluators to counter self-enhancement bias, are also practical steps. Beyond mitigating bias, evaluation must expand to incorporate human-valued traits such as engagement, humor, and empathy. Furthermore, retrieval-augmented evaluation, which integrates external knowledge into the evaluator itself, offers a promising avenue for improving factual consistency, while cross-domain generalization is needed to adapt evaluators to sensitive contexts like medical, legal, or scientific domains.

Finally, the next generation of evaluation tools and techniques will focus on efficiency and advanced reasoning. This includes developing "LLM-as-a-Judge 2.0" systems that incorporate self-consistency and critique-revise loops, and creating interactive, real-time scoring tools (like EvalLM [91]) that enable live feedback. Improving the efficiency of resource-intensive metrics like QuestEval [162] through distillation or pruning, and enhancing evaluator reasoning with methods like Chain-of-Thought and self-consistency remain key priorities. Ultimately, the path forward requires a hybrid approach, combining automated methods with targeted human intervention and establishing rigorous, human-centric protocols for high-stakes domains where automated metrics remain inadequate.

9.3 End to End (Total System)

Evaluating a Retrieval-Augmented Generation (RAG) system as a whole requires moving beyond isolated component metrics to adopt holistic, task-aligned assessments. Future research should prioritize composite evaluation frameworks and balanced metric portfolios that integrate retriever, generator, and end-to-end indicators. Furthermore, evaluation must shift towards more heterogeneous and adaptive benchmarks that incorporate broader task diversity and dynamic requirements like robustness and timeliness, while also being application-aware—prioritizing different metrics based on the specific task.

Several methodological priorities are crucial for advancing end-to-end evaluation. Rigorous cross-model comparisons, complemented by qualitative error analyses, are needed to test framework consistency and reveal where automated metrics diverge from human judgment. A major priority is expanding evaluation to multilingual corpora and specialized domains, and adapting evaluation to new Retrieval-Augmented Generation architectures beyond the simple prepend style, such as Active RAG or iterative retrieval.

The practical deployment of these systems necessitates a strong focus on efficiency, reliability, and human oversight. Future frameworks must benchmark end-to-end systems not only on accuracy but also on latency, computational overhead, and memory cost. A recurring challenge is balancing automation with human judgment; therefore, evaluation pipelines should combine Large Language Model-judges with selective human verification, calibration against domain experts, and structured feedback through interactive dashboards.

Looking forward, the next generation of techniques will likely include retrieval-conditioned metrics that jointly score retrieval and generation. Robustness testing through attribution-aware metrics and contrastive evaluations will become standard, alongside explicit bias mitigation protocols. Ultimately, there is a strong call for community-driven

frameworks, including shared toolkits, low-supervision benchmarks, and standardized pipelines that integrate ethics and privacy metrics into all end-to-end evaluations.

10 Conclusion

The evaluation of Retrieval-Augmented Generation (RAG) systems has rapidly evolved into a complex, multi-faceted discipline, moving far beyond the traditional, isolated metrics of information retrieval and text generation. This survey has provided a comprehensive review of this landscape, systematically charting the diverse approaches for evaluating the retriever, the generator, and the end-to-end system. We have detailed the progression from classical metrics to advanced, semantically-aware frameworks, compiled key benchmark datasets, and analyzed the significant challenges that persist across the entire evaluation pipeline.

Our review reveals a clear trend away from simple, component-wise metrics like ROUGE or Recall@k toward more holistic evaluation paradigms. The rise of automated, end-to-end frameworks such as RAGAS, ARES, and RAGCHECKER, coupled with the widespread adoption of the "LLM-as-a-Judge" approach, exemplifies this shift toward measuring faithfulness, relevance, and context-utilization in an integrated manner. However, these advanced methods are not a panacea. Our analysis of the current challenges highlights persistent issues with evaluator bias, vulnerability to adversarial attacks, a lack of robustness in high-stakes domains, and the high cost and scalability limitations of the most reliable assessment techniques.

Looking ahead, the future of RAG evaluation requires a concerted effort toward creating more reliable, holistic, and practical assessment tools. A primary direction is the development of hybrid evaluation models that combine the scalability of automated judges with the indispensable nuance of human-in-the-loop oversight and expert validation. Future frameworks must integrate not only accuracy but also system-level metrics such as latency and efficiency, and test for robustness against noisy and adversarial conditions. Ultimately, the community must move towards standardized, community-driven benchmarks, robust bias mitigation protocols, and the integration of ethical considerations to ensure that as RAG systems become more powerful, our ability to reliably evaluate them keeps pace.

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